COMP0222 Simultaneous Localisation and Mapping

Lecture 06: Factor Graph Examples

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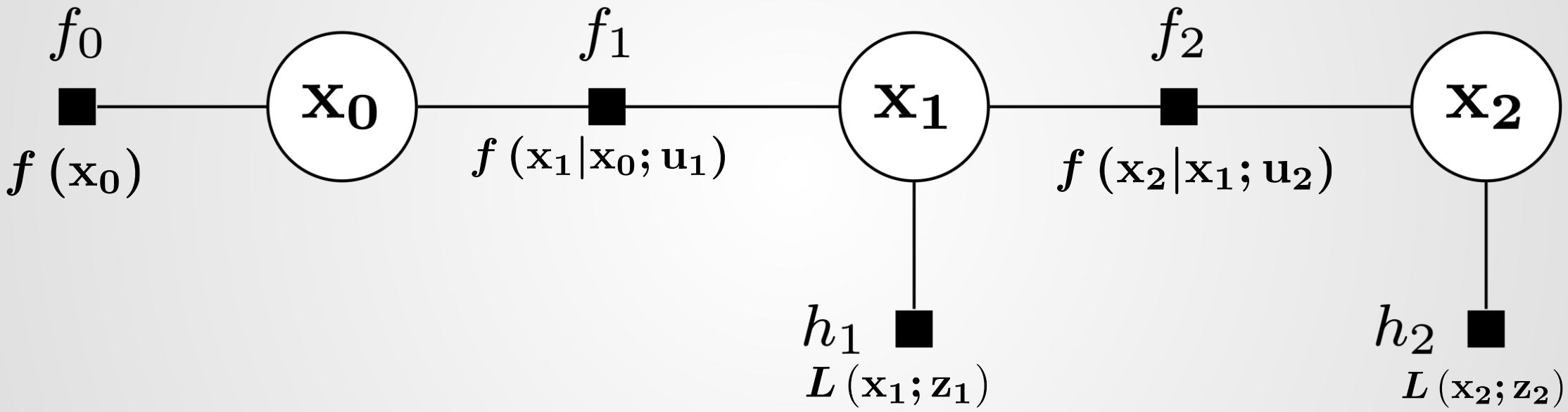
Motivation

- In the last lecture we introduced the factor graph and showed how, for temporal estimation problems, it has the form

$$f\left(\mathbf{x}_{0:k}|\mathbb{I}_k\right) \propto \underline{f\left(\mathbf{x}_0\right)} \prod_{i=1}^{k} \underline{f\left(\mathbf{x}_i|\mathbf{x}_{i-1};\mathbf{u}_i\right)}$$

$$\times \prod_{i=1}^{k} \underline{L\left(\mathbf{x}_i;\mathbf{z}_i\right)}$$

Motivation


f_0
$\mathbf{x_0}$
f_1
$\mathbf{x_1}$
f_2
$\mathbf{x_2}$
$f(\mathbf{x}_0)$
$f(\mathbf{x}_1|\mathbf{x}_0;\mathbf{u}_1)$
$f(\mathbf{x}_2|\mathbf{x}_1;\mathbf{u}_2)$
h_1
$L(\mathbf{x}_1;\mathbf{z}_1)$
h_2
$L(\mathbf{x}_2;\mathbf{z}_2)$

Motivation

- In this lecture, we'll look at a some concrete examples of graphs look like
- We'll build up a few examples to see what the factors look like
- We'll use simple dense sampling the visualize the results

Structure

- Linear Example
- Range-Bearing Measurement Example
- Range-Only Measurement Examples
- Range and Bearing Measurement Example
- Simple Time-Based Estimation Example

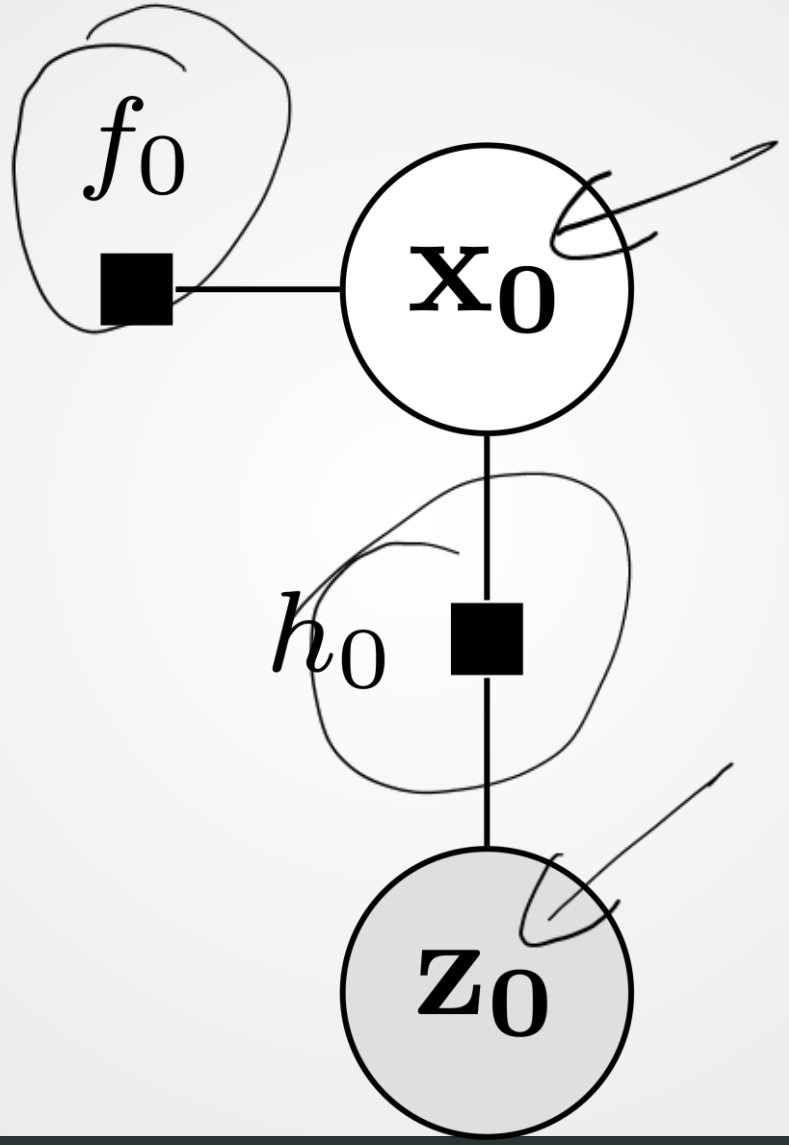
Linear Example

- **Linear Example**
- ***Range-Bearing Measurement Example***
- ***Range-Only Measurement Examples***
- ***Range and Bearing Measurement Examples***
- ***Simple Time-Based Estimation Example***

Linear Example

- We'd like to estimate the 2D position of a platform from a single measurement of its position
- There are no predictions
- We'll arbitrary set all the timesteps to be 0

Linear Example



Factor Graph

- **The factor graph has the form**

$$f(\mathbf{x}_0 | \mathbb{I}_0) \propto f(\mathbf{x}_0)\, L(\mathbf{x}_0; \mathbf{z}_0)$$

Linear 2D Example Vertex State

- **The position of the particle is given by a vertex which contains just its position,**

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Initial Conditions

- **The prior is Gaussian with a given mean and covariance,**

$$\mathbf{x}_0 \sim \mathcal{G}\left(\mathbf{x}_0; \bar{\mathbf{x}}_0, \mathbf{P}_0\right)$$

Initial Conditions Factor

- **Therefore, the factor which encodes this information is**

$$f(\mathbf{x}_0) \propto \underbrace{\exp\left\{-\frac{1}{2}(\mathbf{x}_0 - \bar{\mathbf{x}}_0)^\top \mathbf{P}_0^{-1}(\mathbf{x}_0 - \bar{\mathbf{x}}_0)\right\}}_{L(x_0)}$$

Initial Conditions Factor

- For this example, let's say

$$\mathbf{x}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

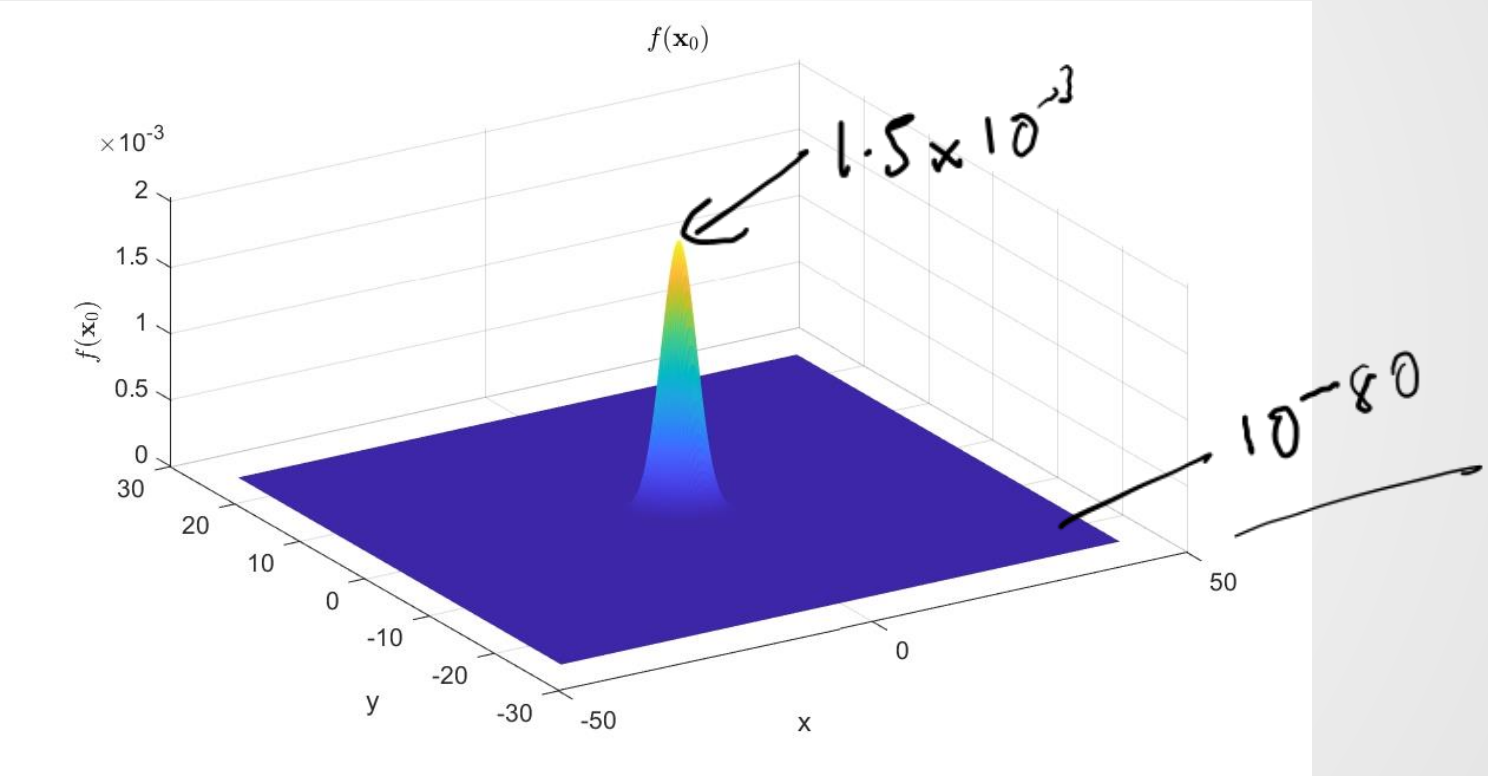
$$\mathbf{P}_{0|0} = \begin{bmatrix} 0.125 & 0.100 \\ 0.100 & 0.500 \end{bmatrix}^{-1}$$

Visualizing the Prior Factor

- **We can visualize this factor by densely varying the value of $\mathbf{x}_o$ over a grid**

$$f(x_0) \rightarrow \begin{pmatrix} x \\ y \end{pmatrix}$$

Prior

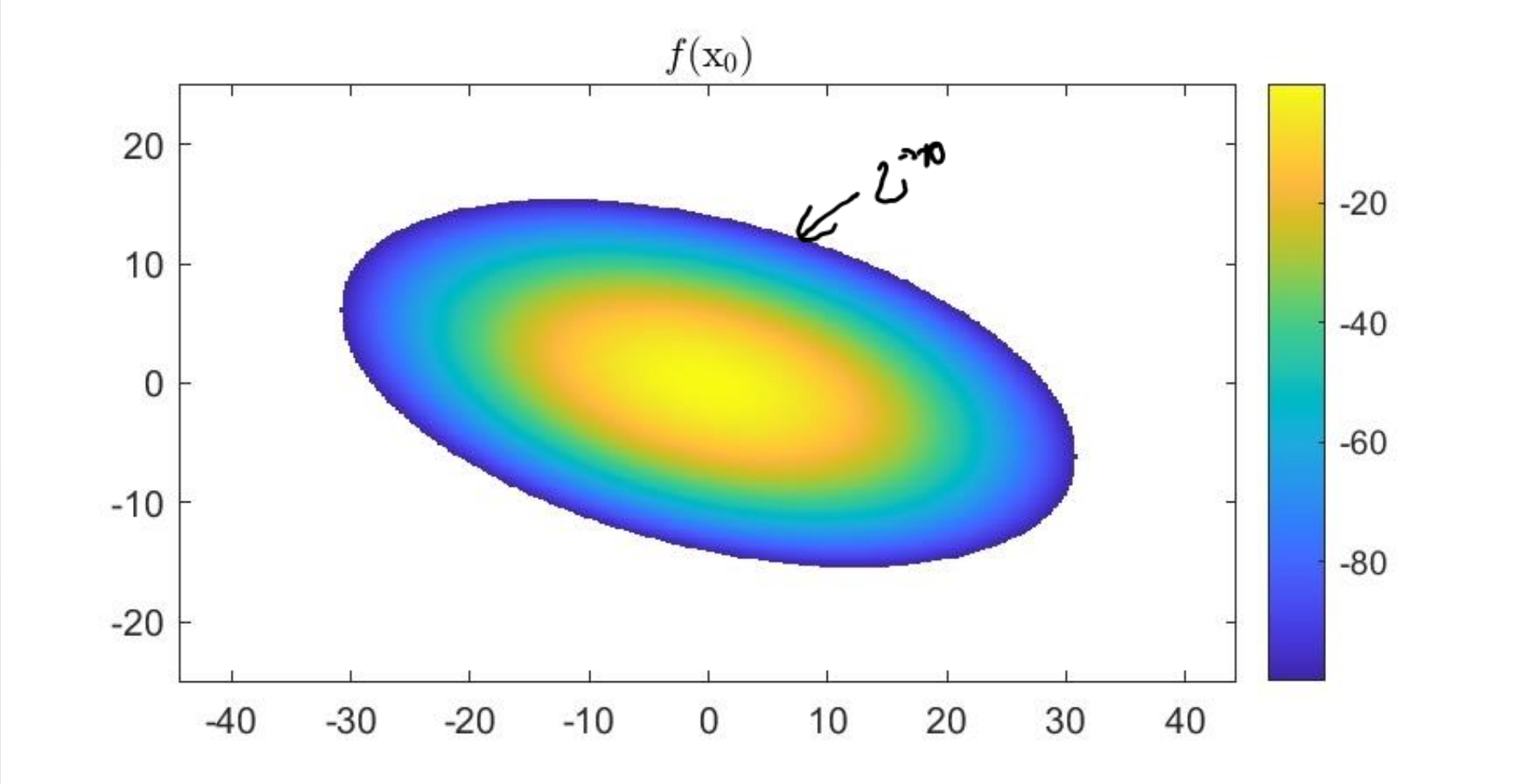

$f(\mathbf{x}_0)$
$\times 10^{-3}$
2
1.5
1
0.5
0
$f(\mathbf{x}_0)$
30
20
10
0
-10
-20
-30
y
-50
0
50
x
1.5×10^{-3}
10^{-80}

Visualizing the Prior Factor

- We can visualize this factor by densely varying the value of $\mathbf{x}_o$ over a grid
- The magnitude of the PDF can vary substantially
- Therefore, instead we plot the *log* to flatten the dynamic range
- We also truncate the values so we don't show anything smaller than -100

$a > b > 0$ $\quad$ $\ln a > \ln b$

Visualizing the Prior



Observation Model

- **The system directly observes the platform position plus some additive observation noise,**

$$\mathbf{z}_k = \begin{pmatrix} z_k^x \\ z_k^y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \mathbf{w}_k$$

Likelihood Factor

- **The likelihood factor looks very similar to the prior factor,**

$$L\left(\mathbf{x}_0;\mathbf{z}_0\right) \propto \exp\left\{-\frac{1}{2}\left(\mathbf{z}_0-\mathbf{x}_0\right)^\top \mathbf{R}_0^{-1}\left(\mathbf{z}_0-\mathbf{x}_0\right)\right\}$$

$f(z_0|x_0)$

Observation Values

- Let's arbitrarily say

$$\mathbf{z}_0 = \begin{bmatrix} -20 \\ -5 \end{bmatrix}$$

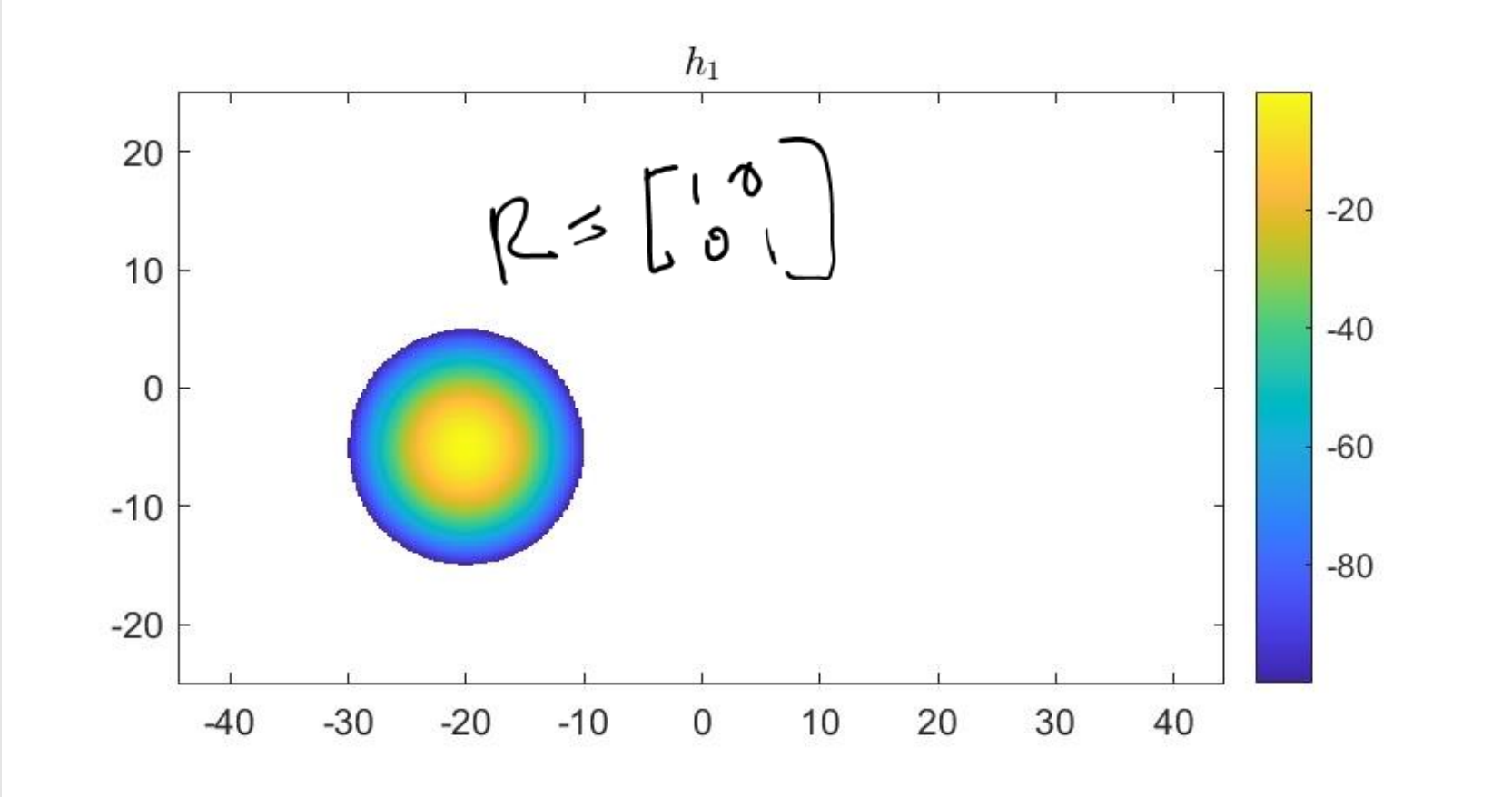
$$\mathbf{R}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Visualizing the Likelihood

- **We plot the likelihood by holding z constant and vary x**

$$L(z; x_i)$$

Visualizing the Likelihood

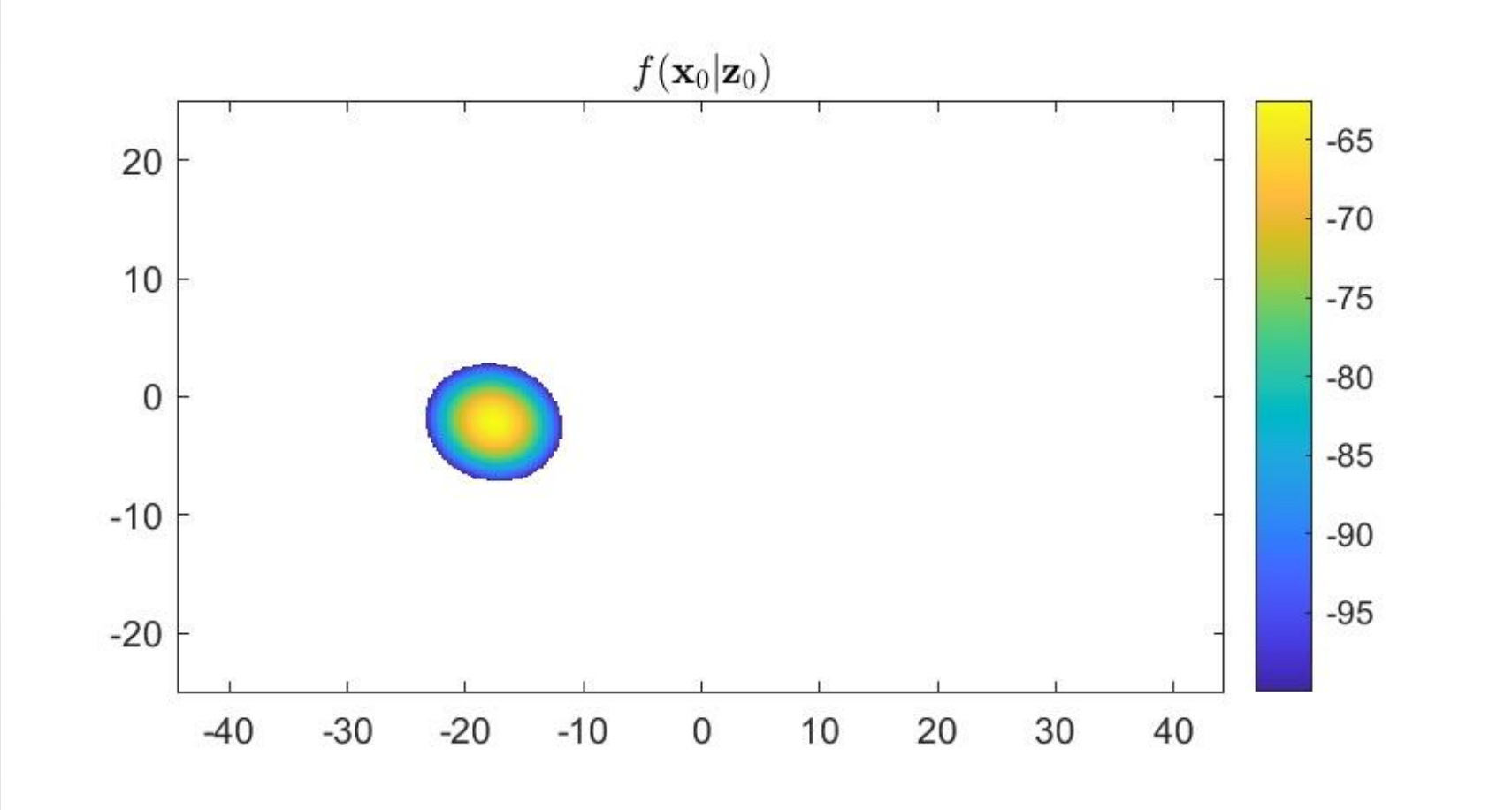


Visualizing the Full Factor Graph Output

- We compute the product of the prior and observation factors and take logs
- We can compute the same result by adding the logs of each term

$$\ln f(x_0; \Psi_w) = \ln f(x_0) + \ln L(z_0; x_0)$$

Visualizing the Joint Distribution



Visualizing the Full Factor Graph Output

- The result is a distribution with a single point
- This might seem surprising, but recall we have linear models and Gaussian noises, and we are multiplying these terms together
- The results are always guaranteed to be a Gaussian

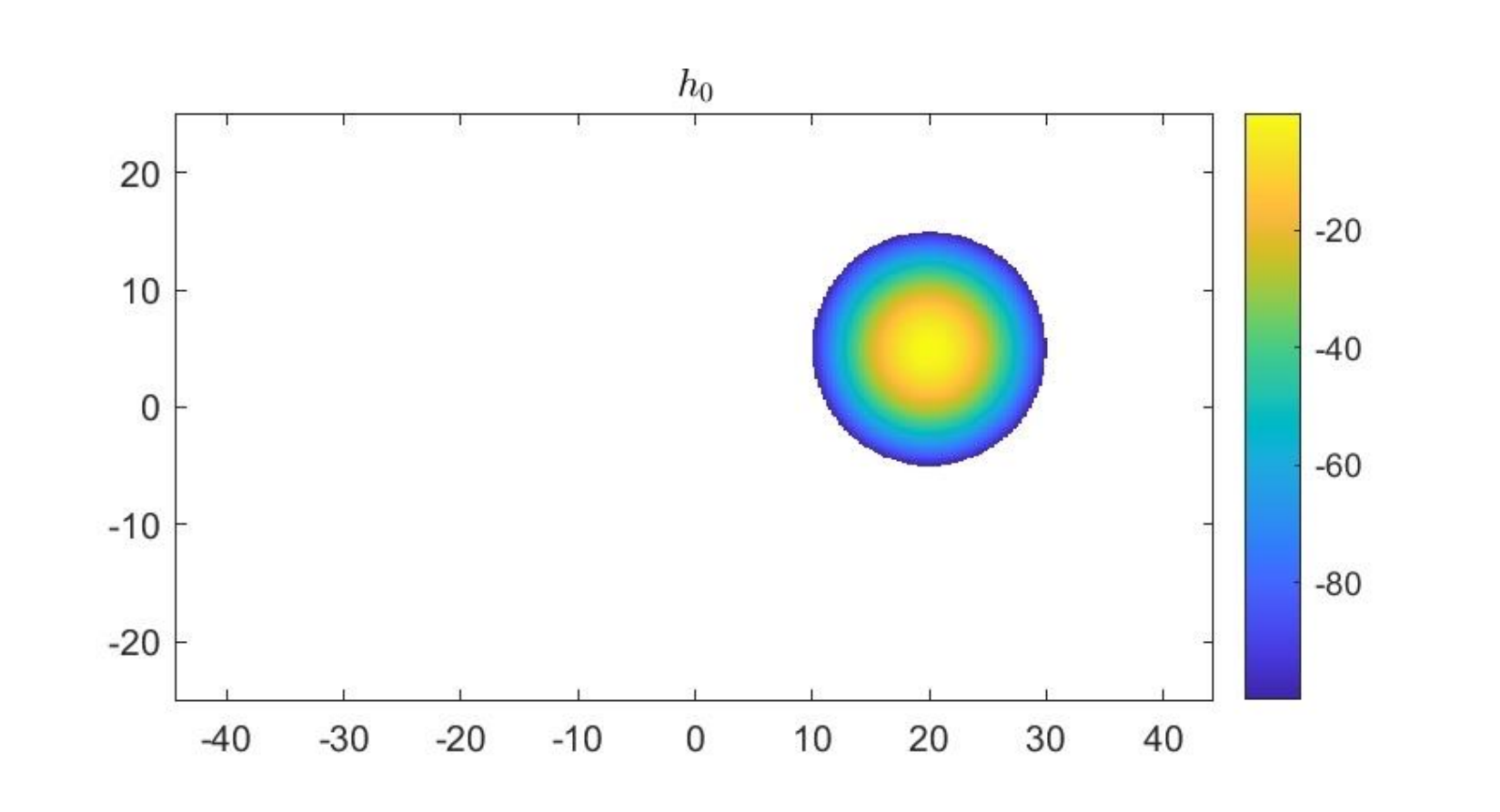
More Measurements

- Suppose the measurement is now

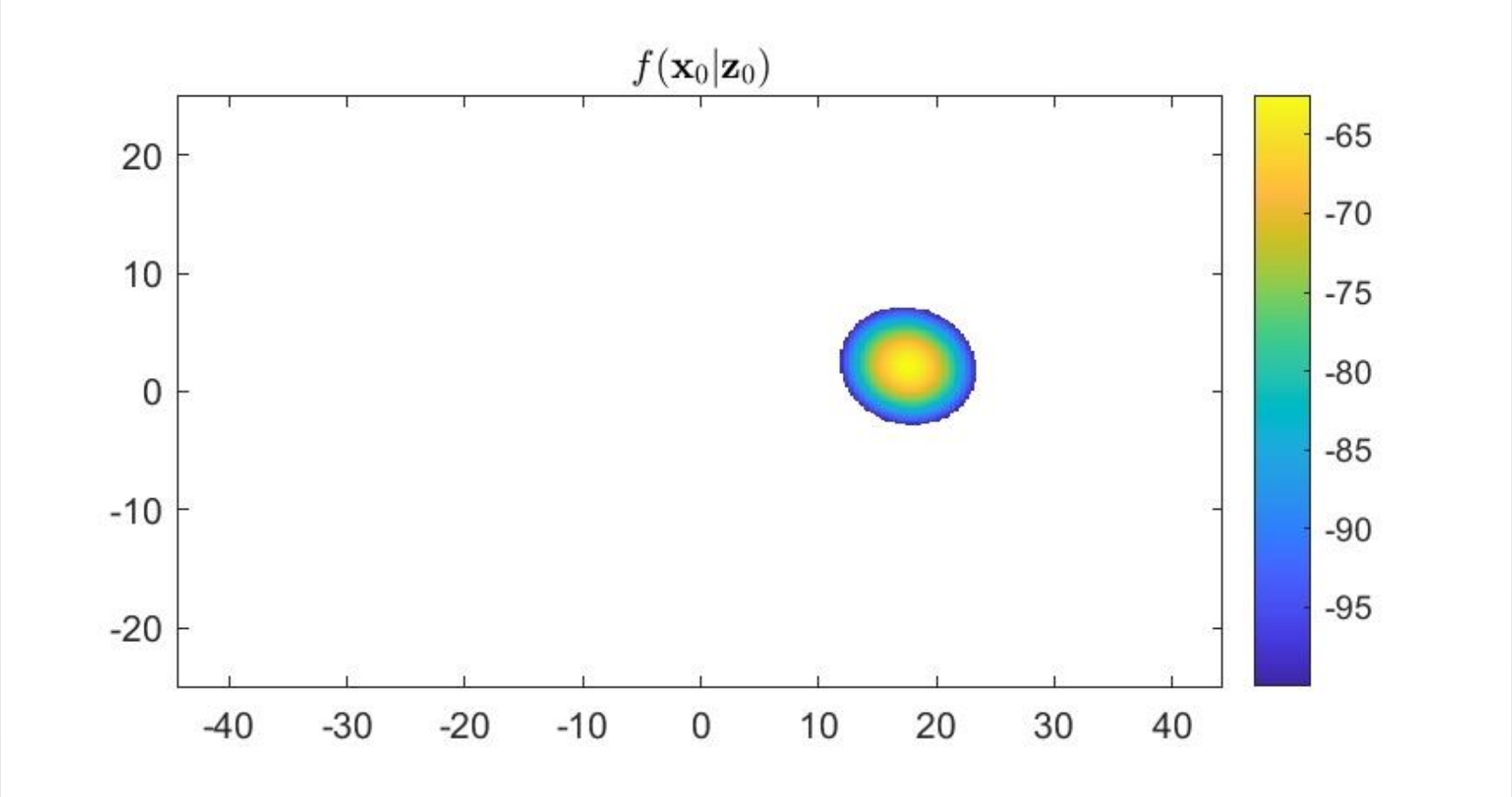
$$\mathbf{z}_0 = \begin{bmatrix} 20 \\ 5 \end{bmatrix}$$

- The prior stays the same, but the likelihood and the factor graph estimate change

Visualizing the Likelihood



Visualizing the Joint Distribution



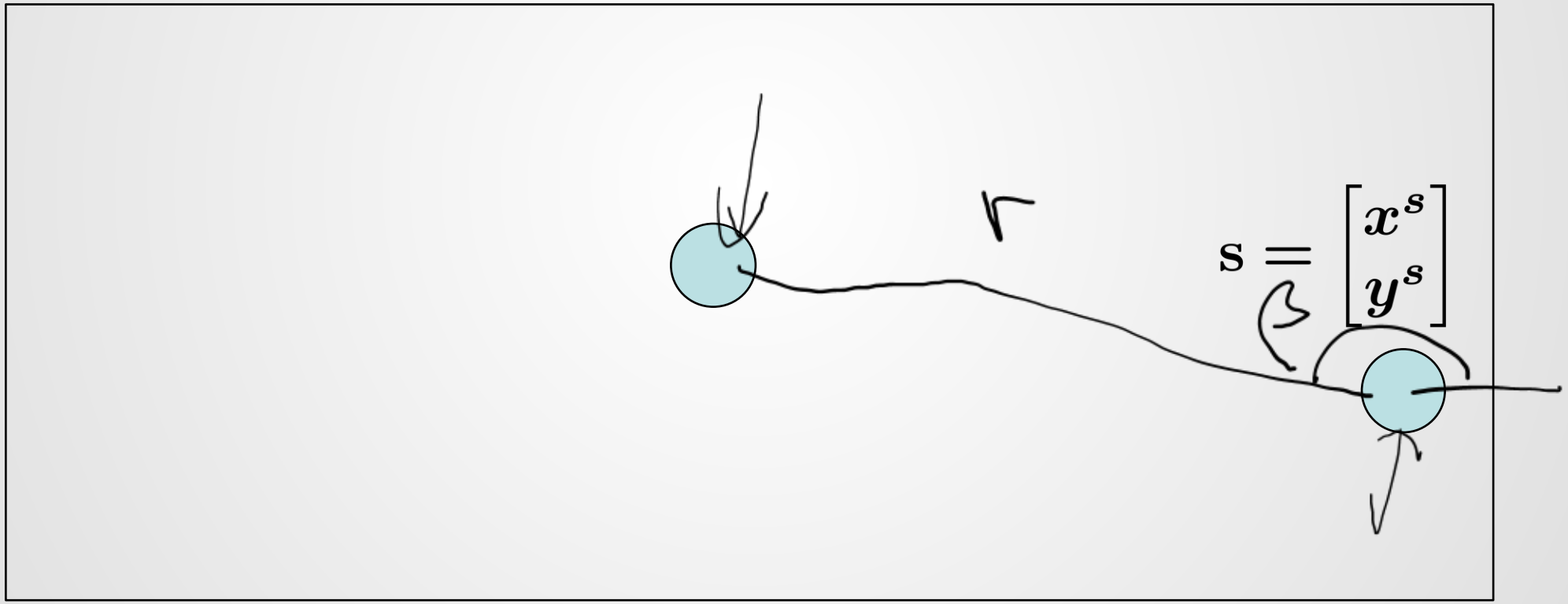
Range-Bearing Measurement Example

- *Linear Example*
- **Range-Bearing Measurement Example**
- *Range-Only Measurement Examples*
- *Range and Bearing Measurement Examples*
- *Simple Time-Based Estimation Example*

Range-Bearing Measurement Example

- Now let's look at a more interesting problem
- We are going to swap the sensor for one which returns the *range* and *bearing* to the platform
- These measurements are taken from a second sensing platform
- The prior factor is the same as before, so we only need to look at the measurement likelihood

Range-Bearing Measurement Example



Range-Bearing Observation Model

- **The observation model is nonlinear and is of the form**

$$\mathbf{z}_0 = \begin{bmatrix} r_0 \\ \beta_0 \end{bmatrix}$$

$$\mathbf{z}_0 = \mathbf{h}\left[\mathbf{x}_0, \mathbf{s}\right] + \mathbf{w}_0$$

Range-Bearing Observation Model

- **Define the values**

$$\Delta x_0^s = -(x^s - x_0)$$
$$\Delta y_0^s = -(y^s - y_0)$$

Observation Model

- **Therefore, the observation model is**

$$r_0 = \sqrt{\left(\Delta x_0^s\right)^2 + \left(\Delta y_0^s\right)^2} + \mathbf{w}_0^r$$

$$\beta_0 = \arctan\left(\frac{\Delta y_0^s}{\Delta x_0^s}\right) + \mathbf{w}_0^\beta$$

Observation Factor

$f(z_0 | x_0)$

- We can use our Gaussian form again
- However, to simplify the notation, we'll introduce the *error function*
- This is the error between the measured and predicted observation,

$$\mathbf{l}\left[\mathbf{z}_0, \mathbf{x}_0; \mathbf{s}\right] = \mathbf{z}_0 - \mathbf{h}\left[\mathbf{x}_0; \mathbf{s}\right] = w_0$$

Error Functions

- **Therefore, the likelihood factor has the form**

$$L\left(\mathbf{x}_0;\mathbf{z}_0\right) \propto \exp\left\{-\frac{1}{2}\mathbf{l}\left[\mathbf{z}_0,\mathbf{x}_0\right]^\top \mathbf{R}_0^{-1}\mathbf{l}\left[\mathbf{z}_0,\mathbf{x}_0\right]\right\}$$

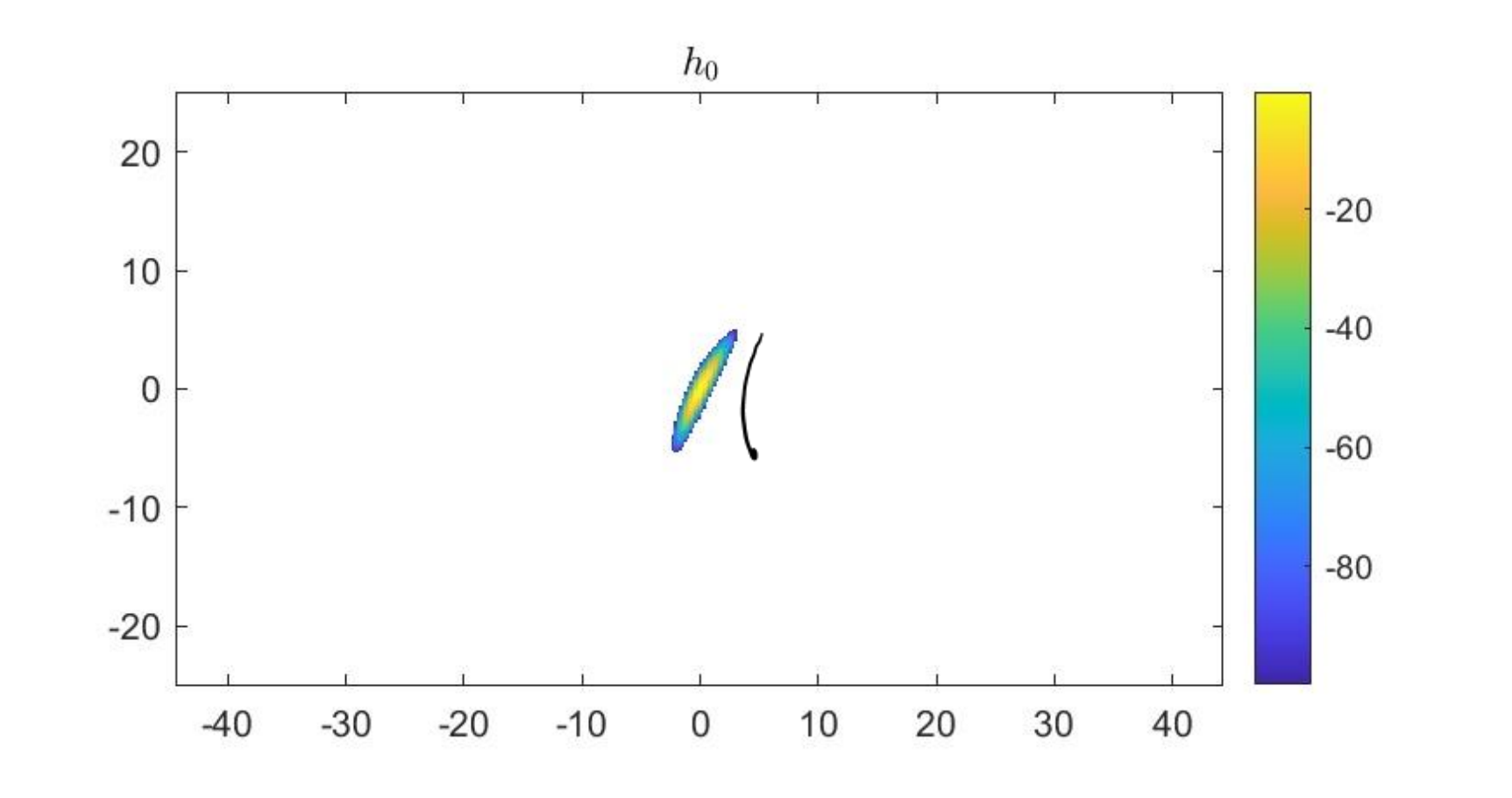
z→均值

Visualizing the Sensor Likelihood

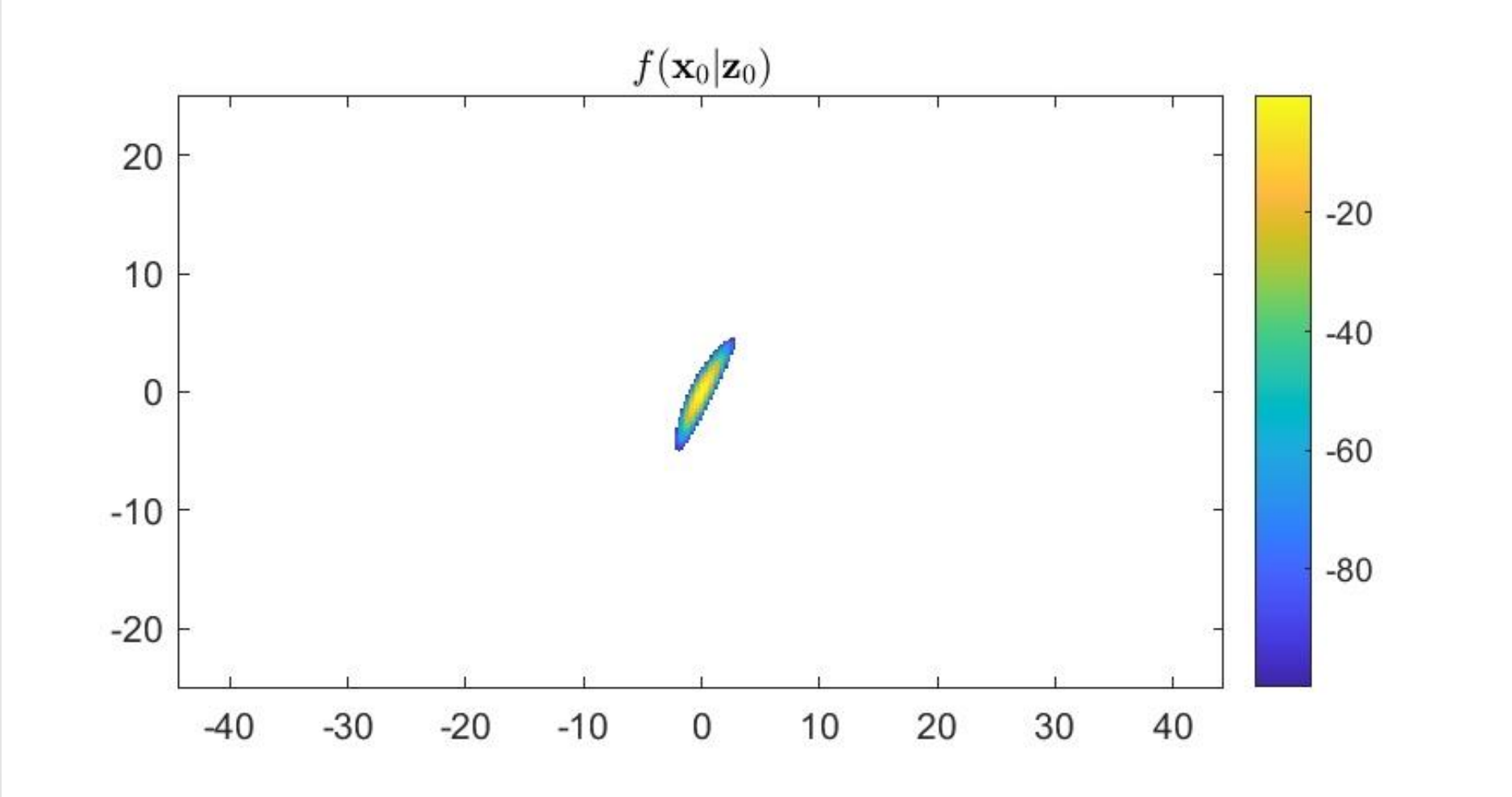
- As before, we'll compute the likelihood by fixing $\mathbf{z}$ and varying $\mathbf{x}_o$
- The sensor platform position and covariances are

$$\mathbf{s} = \begin{bmatrix} 30 \\ -15 \end{bmatrix}, \mathbf{R}_0 = \begin{bmatrix} 0.01 & 0 \\ 0 & (\pi/180)^2 \end{bmatrix}$$

Visualizing the Likelihood



Visualizing the Fused Result

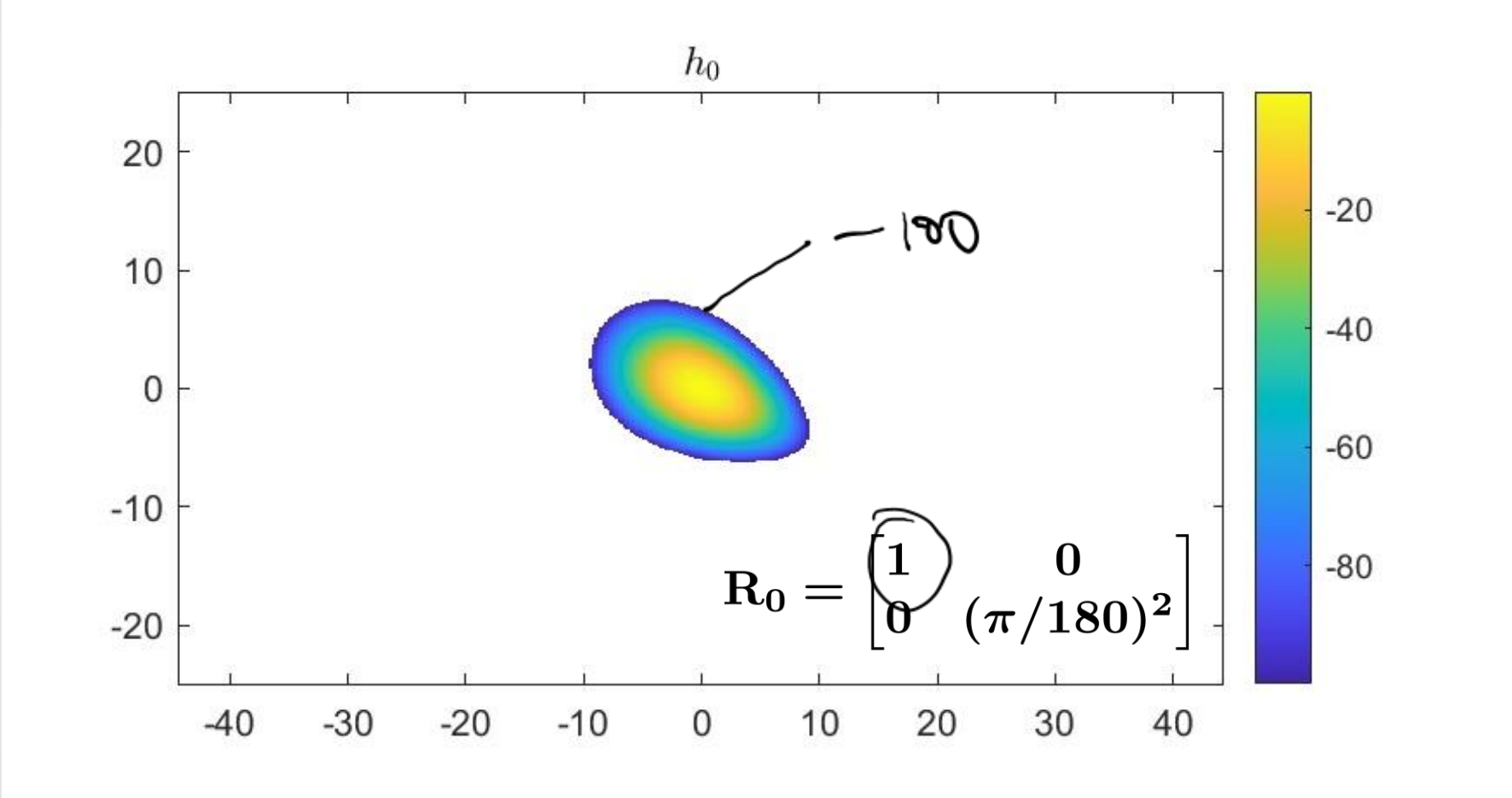


More Results

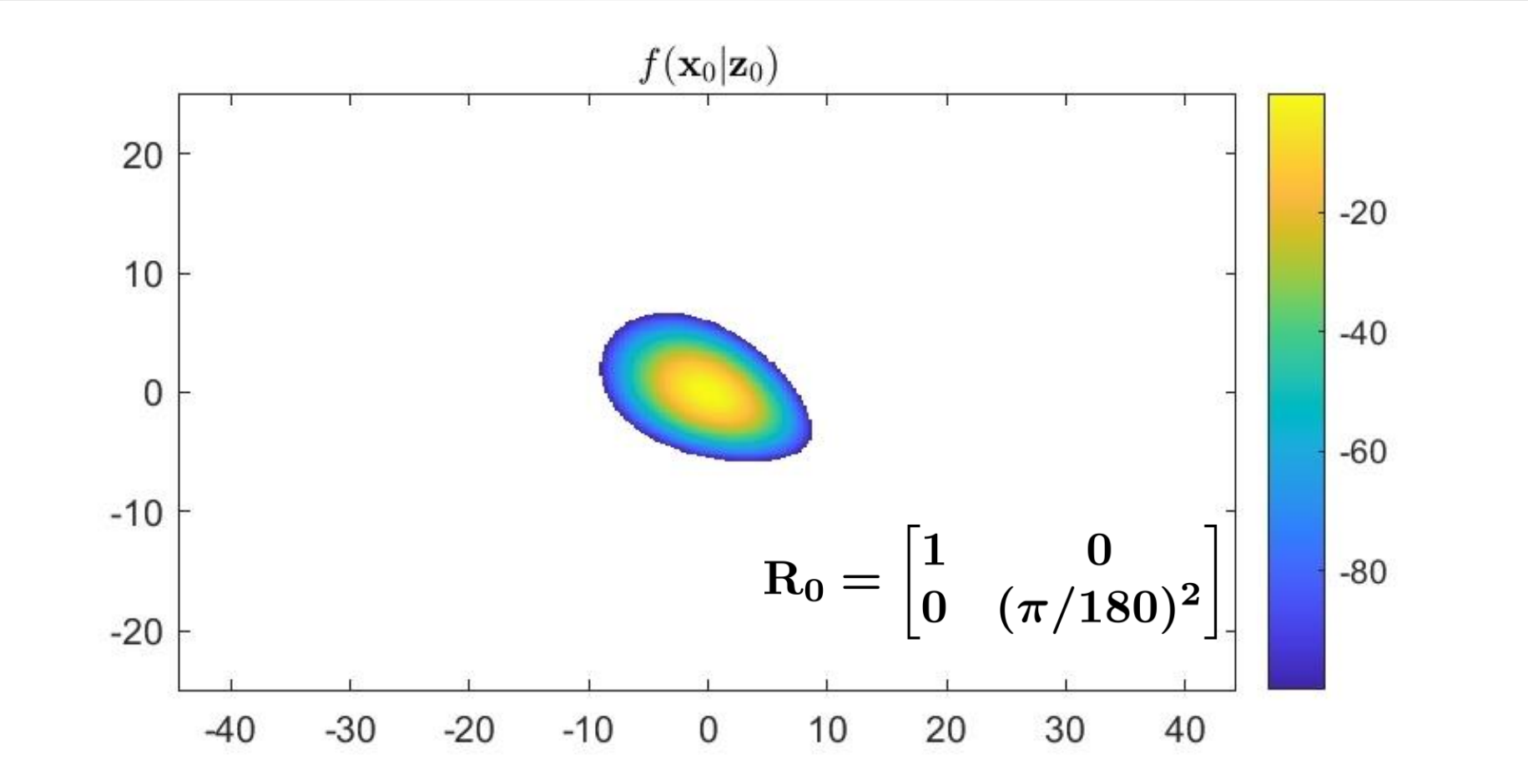
- You can see here that the non-Gaussian nature is evident – you have a probability contour that's folded a little along a line of constant radius

- We can vary the noise parameters to explore this a bit more

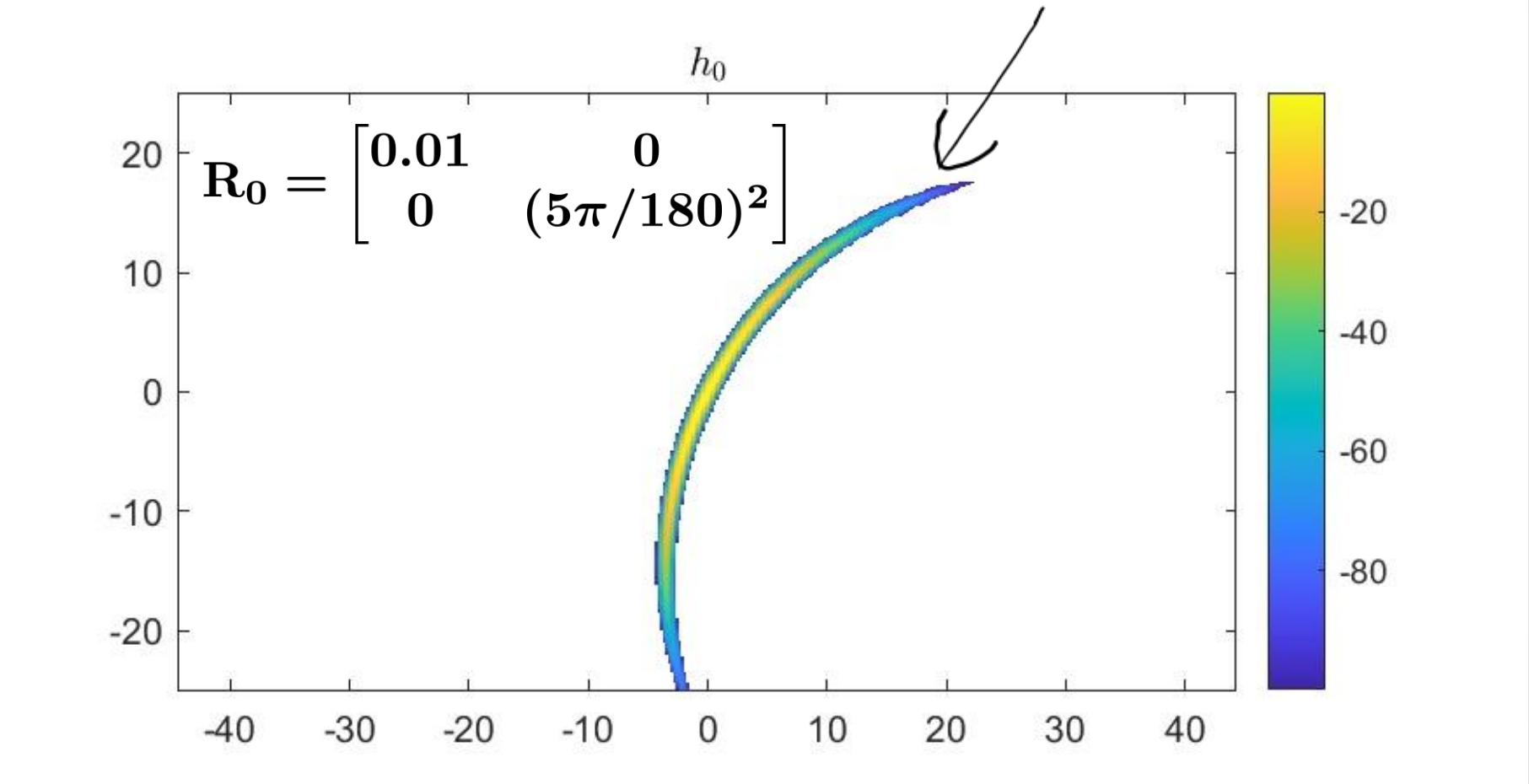
Visualizing the Likelihood



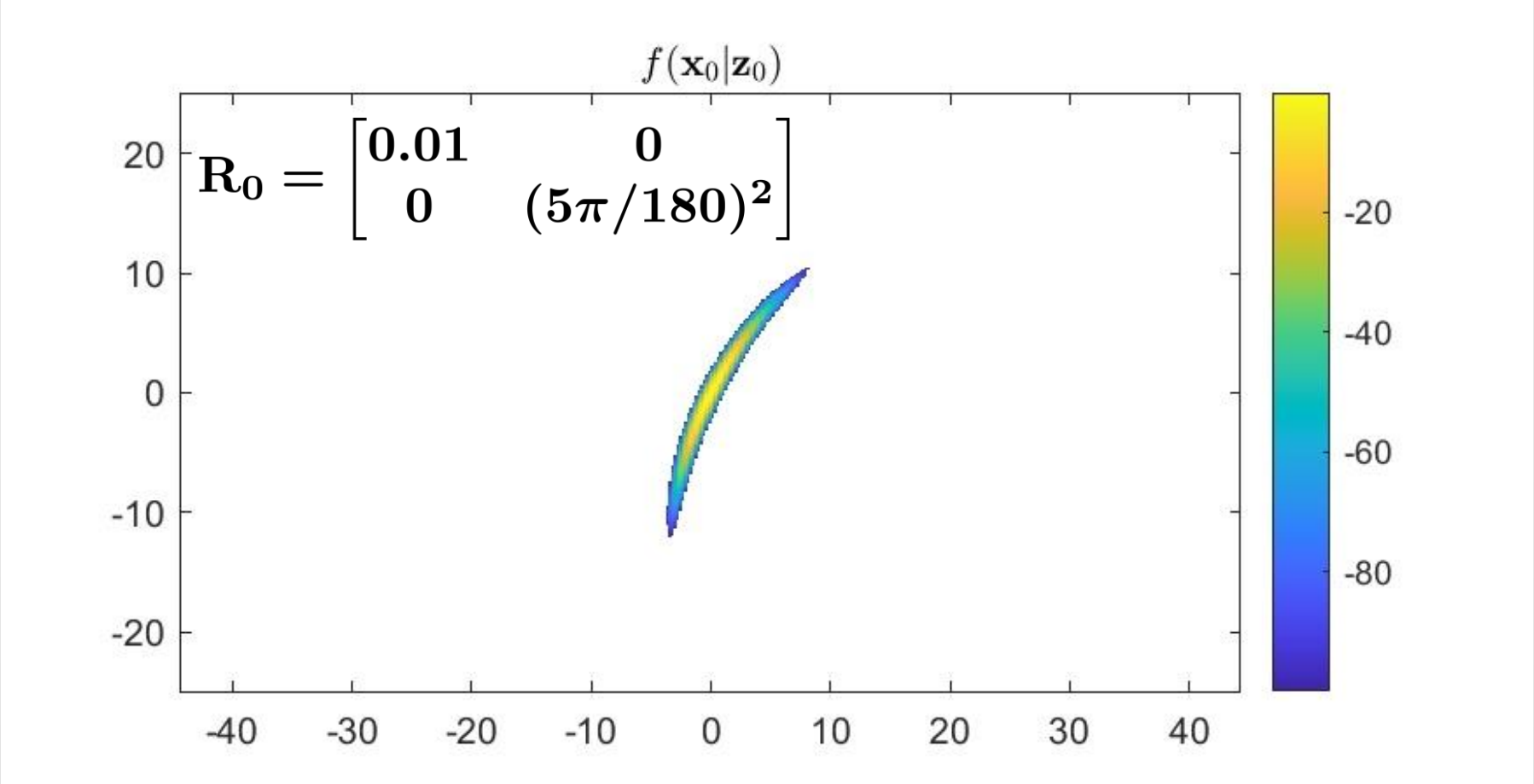
Visualizing the Fused Result



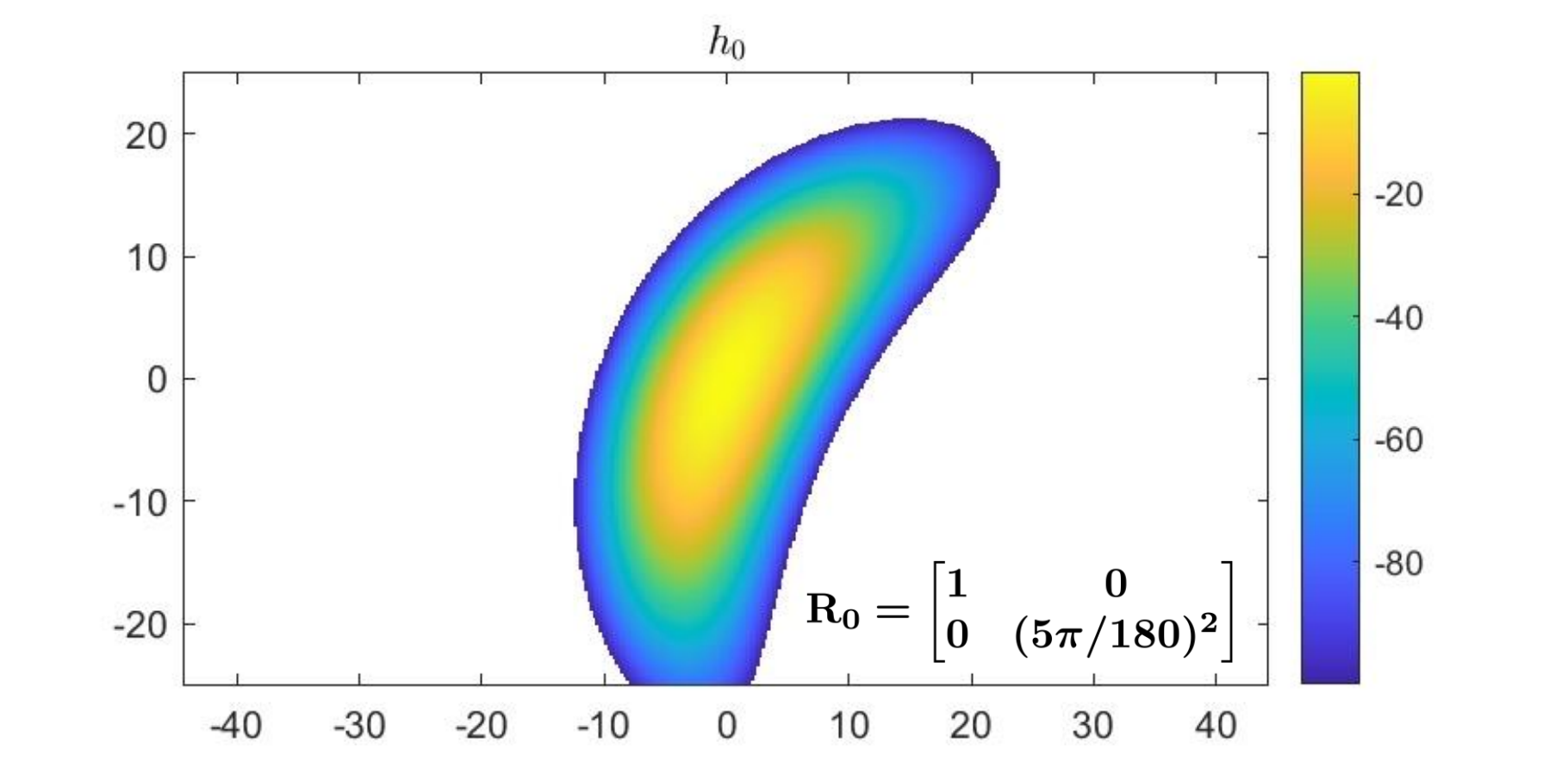
Visualizing the Likelihood



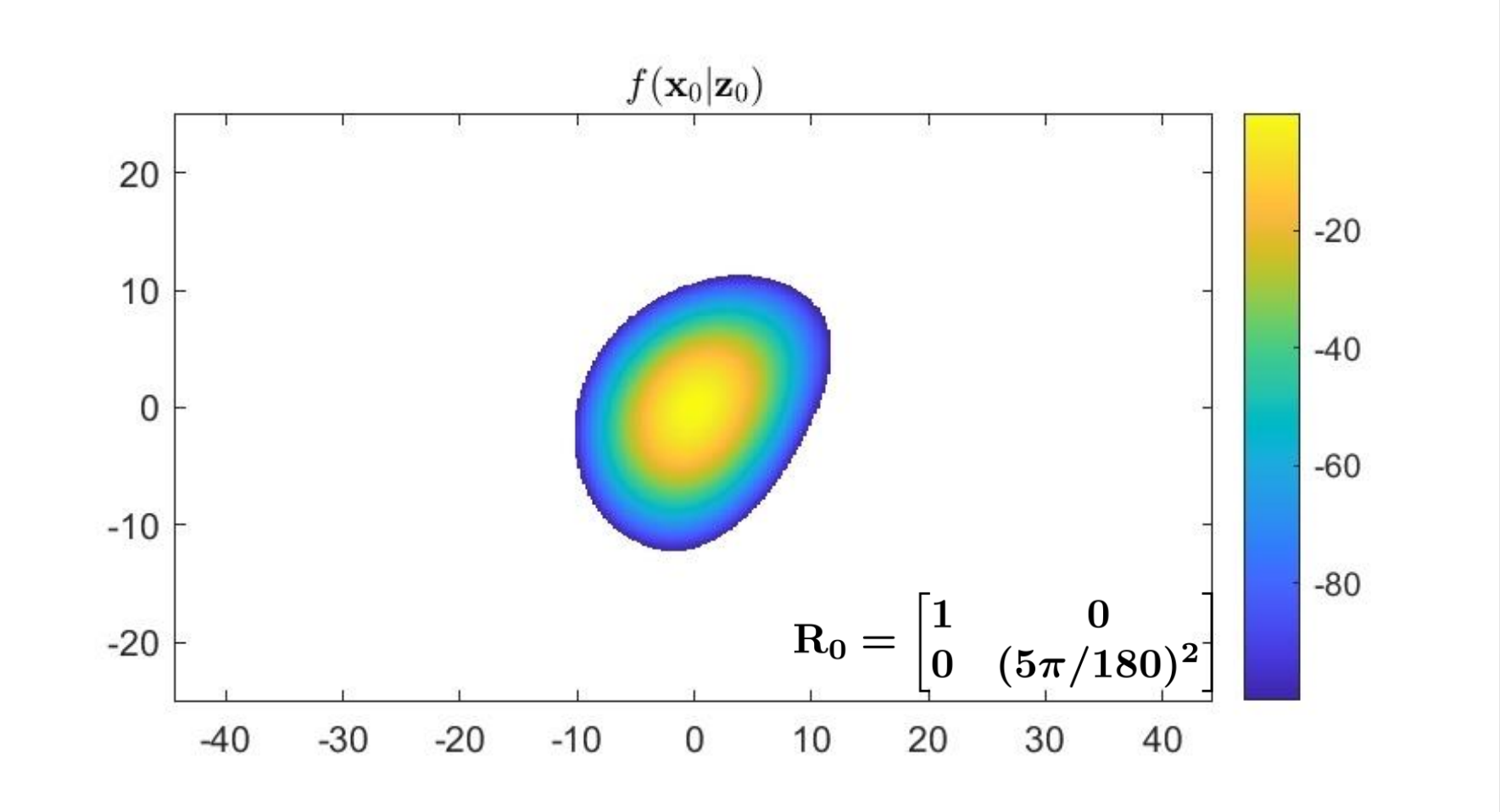
Visualizing the Fused Result



Visualizing the Likelihood



Visualizing the Fused Result



Gaussian Noises, Non-Gaussian Likelihoods

$w \sim G$

- We said the observation noise is Gaussian
- However, the likelihood function is *not* Gaussian, especially for very large noise values
- The fused update isn't Gaussian either
- Why isn't this the case?

Gaussian Noises, Non-Gaussian Likelihoods

- **The reason is the *nonlinearity* of the error function,**

$$\mathbf{l}\left[\mathbf{z}_0, \mathbf{x}_0; \mathbf{s}\right] = \mathbf{z}_0 - \mathbf{h}\left[\mathbf{x}_0; \mathbf{s}\right]$$

$$\exp\left(-\frac{1}{2} l^T P^{-1} l\right)$$

Conditional Probabilities vs. Likelihoods

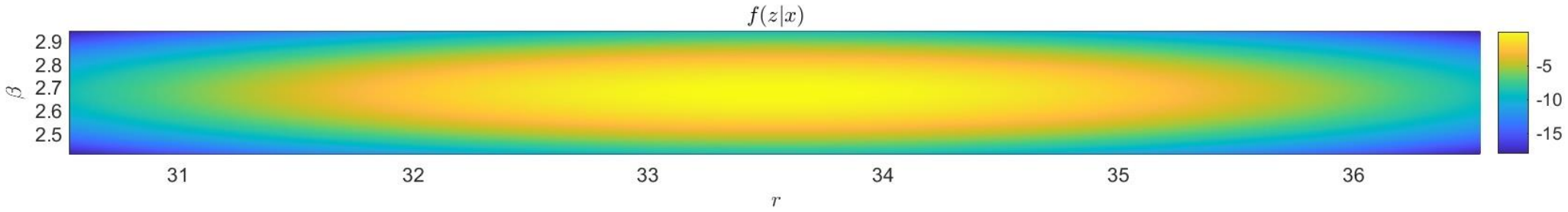
- We can also use this likelihood function to demonstrate the difference between conditional probability and likelihood

- Recall that, with a likelihood, we vary x and keep z constant
- However, we can keep x fixed, vary z and compute the value of the likelihood factor

$f(z|x)$

$p(x|z)$

Holding x fixed, varying z



Range-Only Measurement Examples

- *Linear Example*
- *Range-Bearing Measurement Example*
- Range-Only Measurement Examples
- *Range and Bearing Measurement Example*
- *Simple Time-Based Estimation Example*

Range-Only Measurement Examples

- Consider the case that we can only estimate the range to the platform from a sensor

- This is different from the previous cases in that we have 1 measurement and 2 degrees of freedom

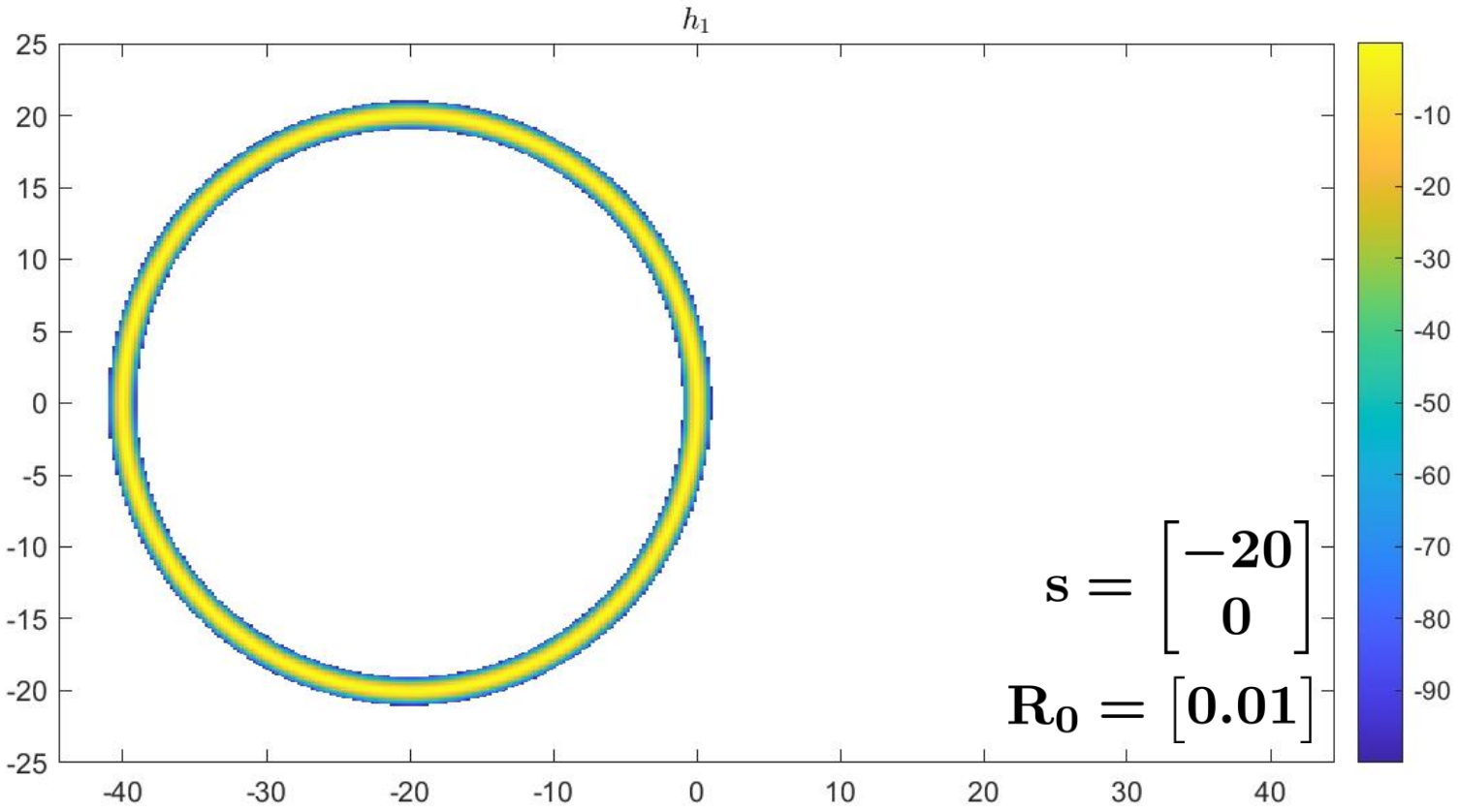
Range-Only Measurement Examples

- This is obtained directly from the range-bearing model,

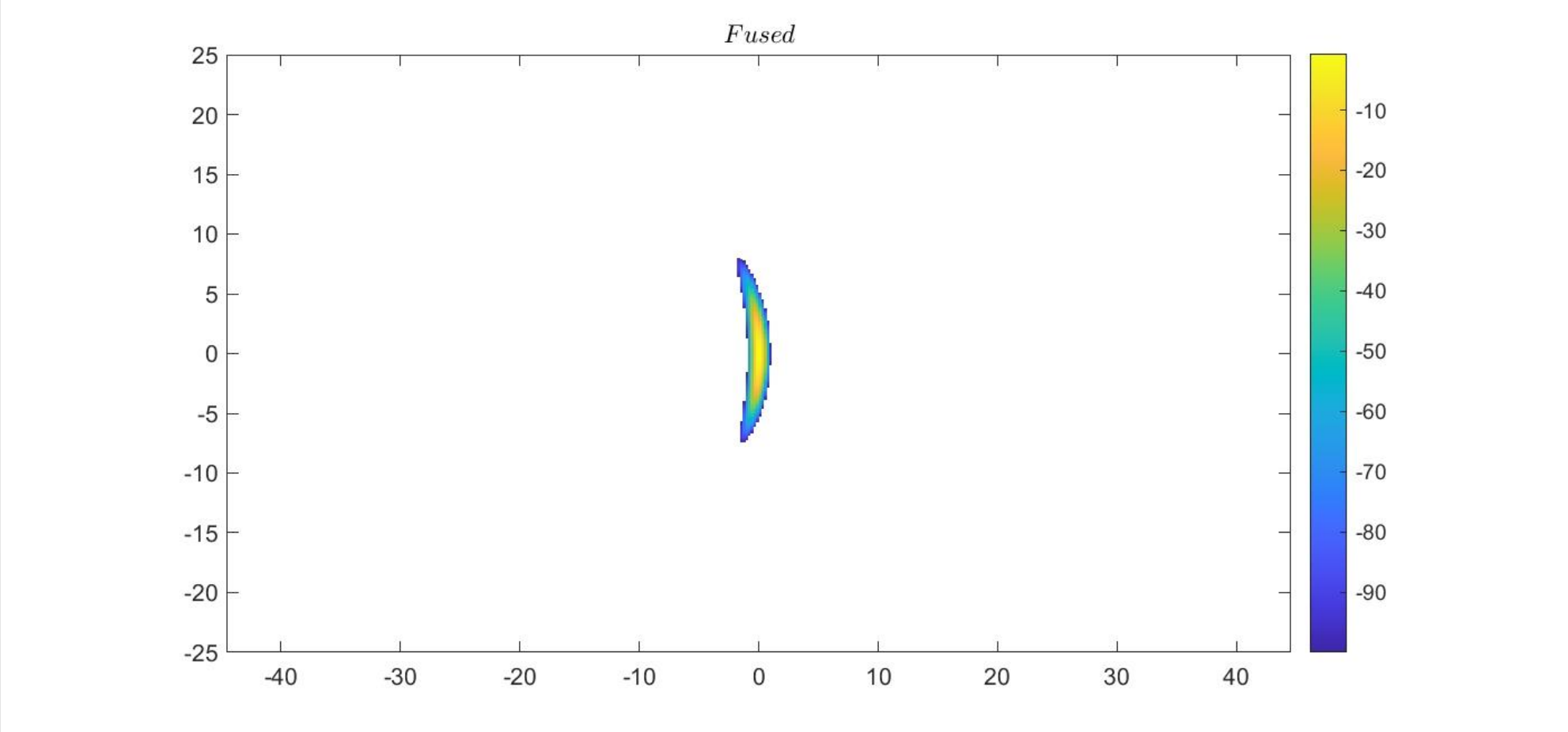
$$r_0 = \sqrt{\left(\Delta x_0^s\right)^2 + \left(\Delta y_0^s\right)^2} + \mathbf{w}_0^r$$

- Similarly, we just extract the first row of the error model and the first element of the observation noise covariance matrix

Range-Only Measurement Likelihood



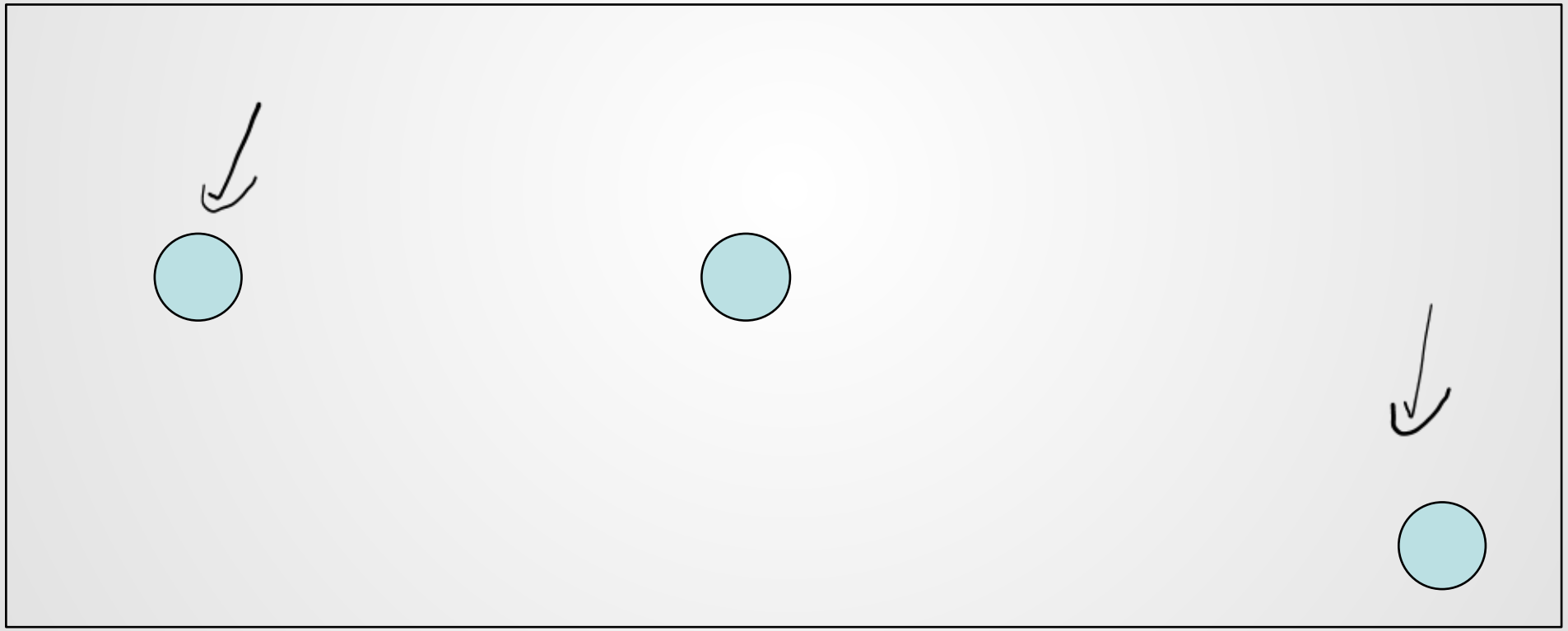
Fused Result



Effect of the Prior

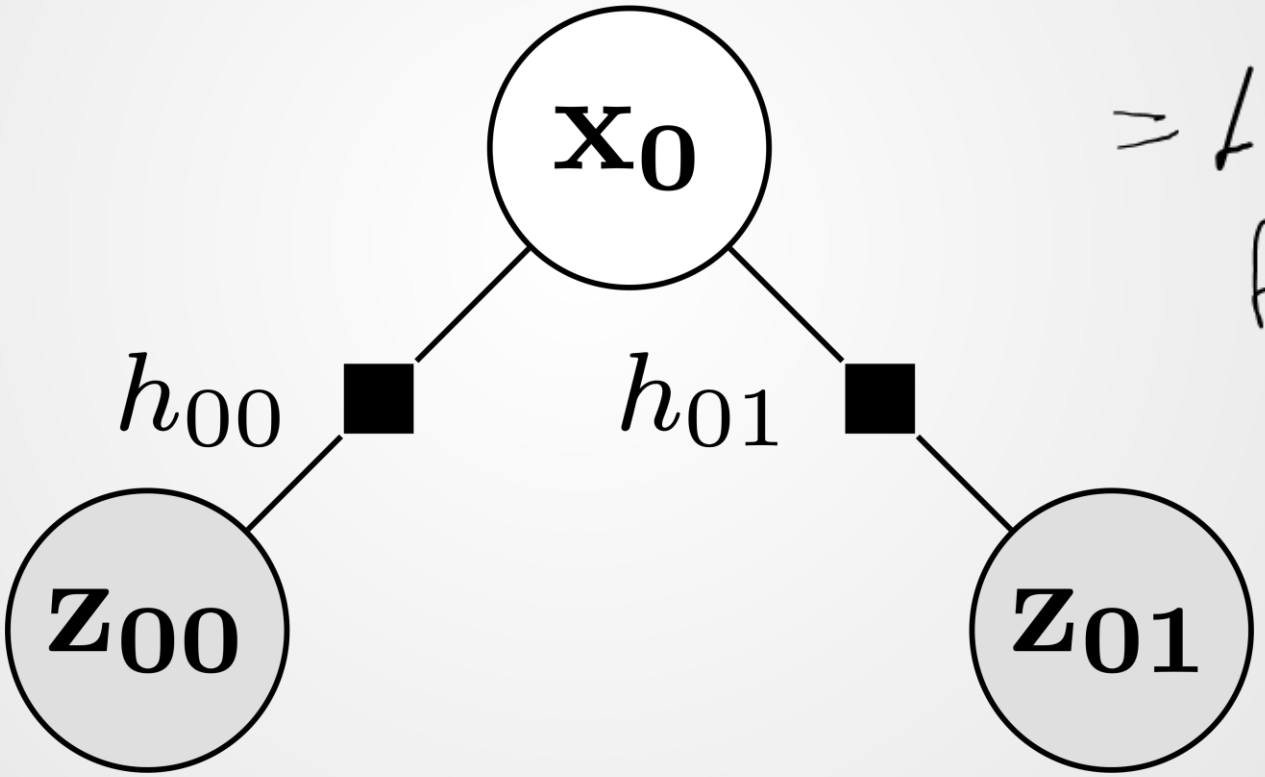
- In this case, the prior has a significant effect on the estimate and constrains its location
- Therefore, this shows how it can be important
- However, we can also get away with a graph without a direct prior, if the state is observable from the sensors
- We can illustrate this through the use of two range-only sensors

Multiple Range Measurement Example



Multiple Range Measurement Example

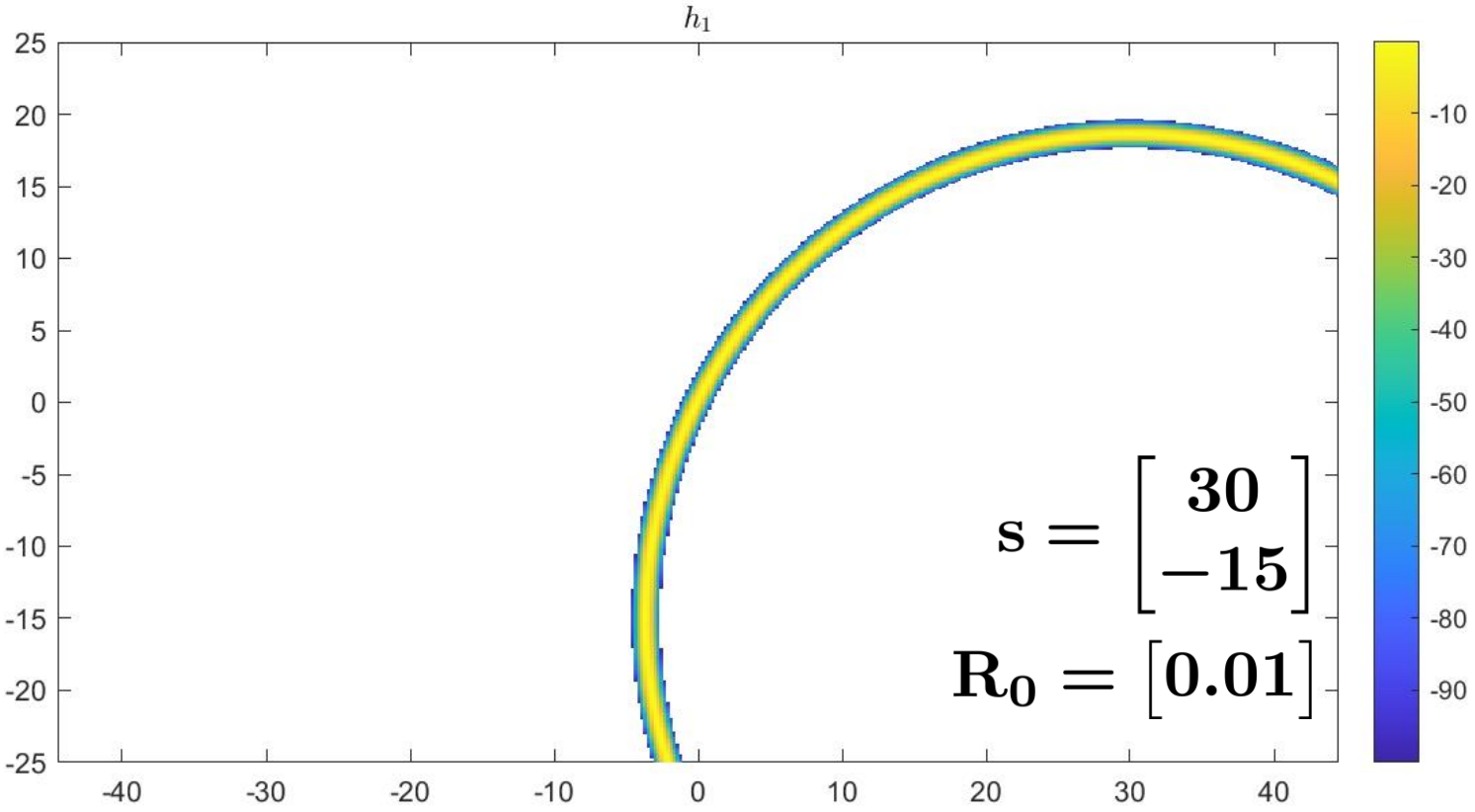
$$f(z_1, z_2 | x) = f(z_1 | x) \, f(z_2 | x)$$

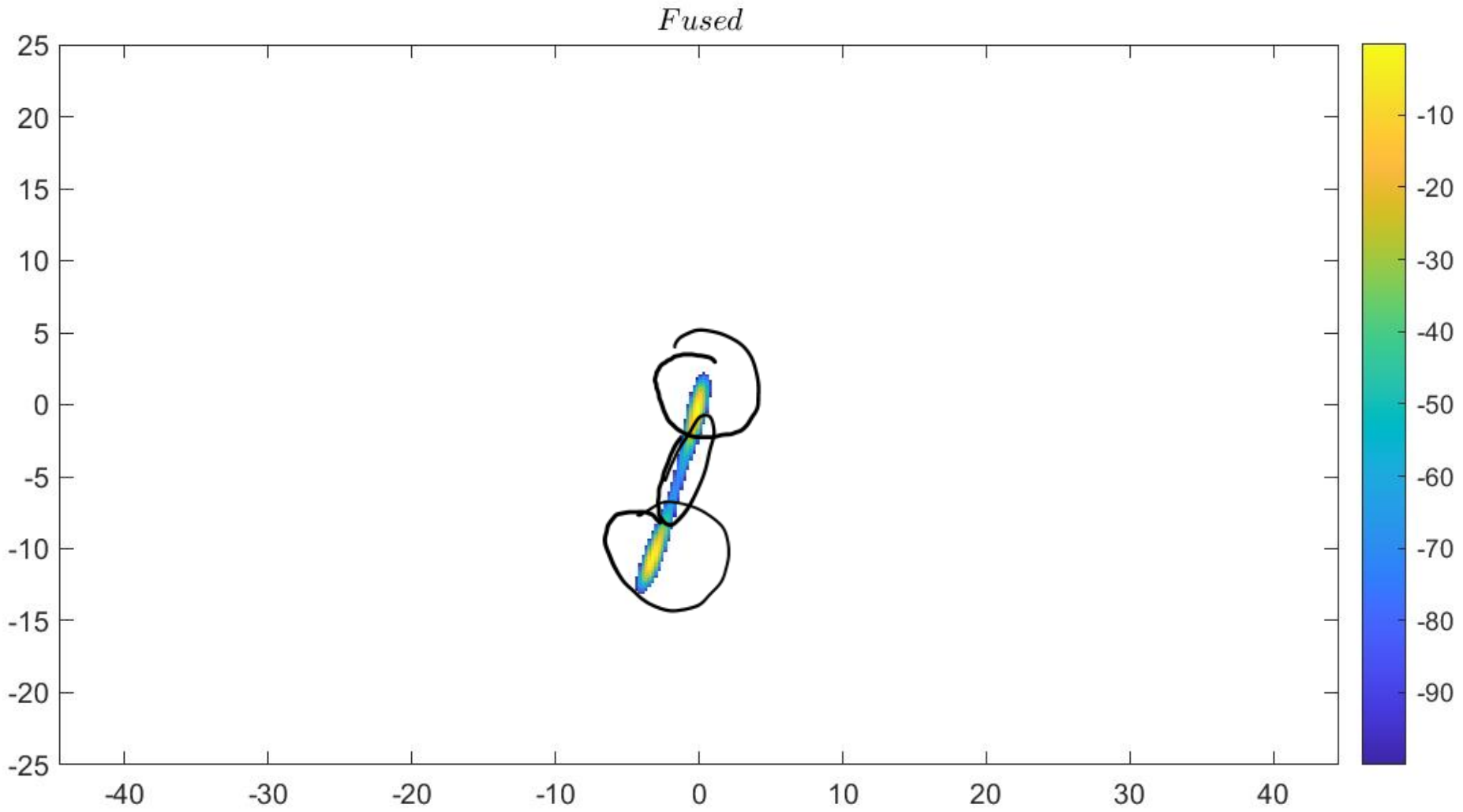


Range-Only Measurement Factor

- The expression for the second factor has the same form as the first
  - We just substitute the new sensor location and the new observation

Second Range-Only Sensor Likelihood




Fused
25
20
15
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-70
-80
-90

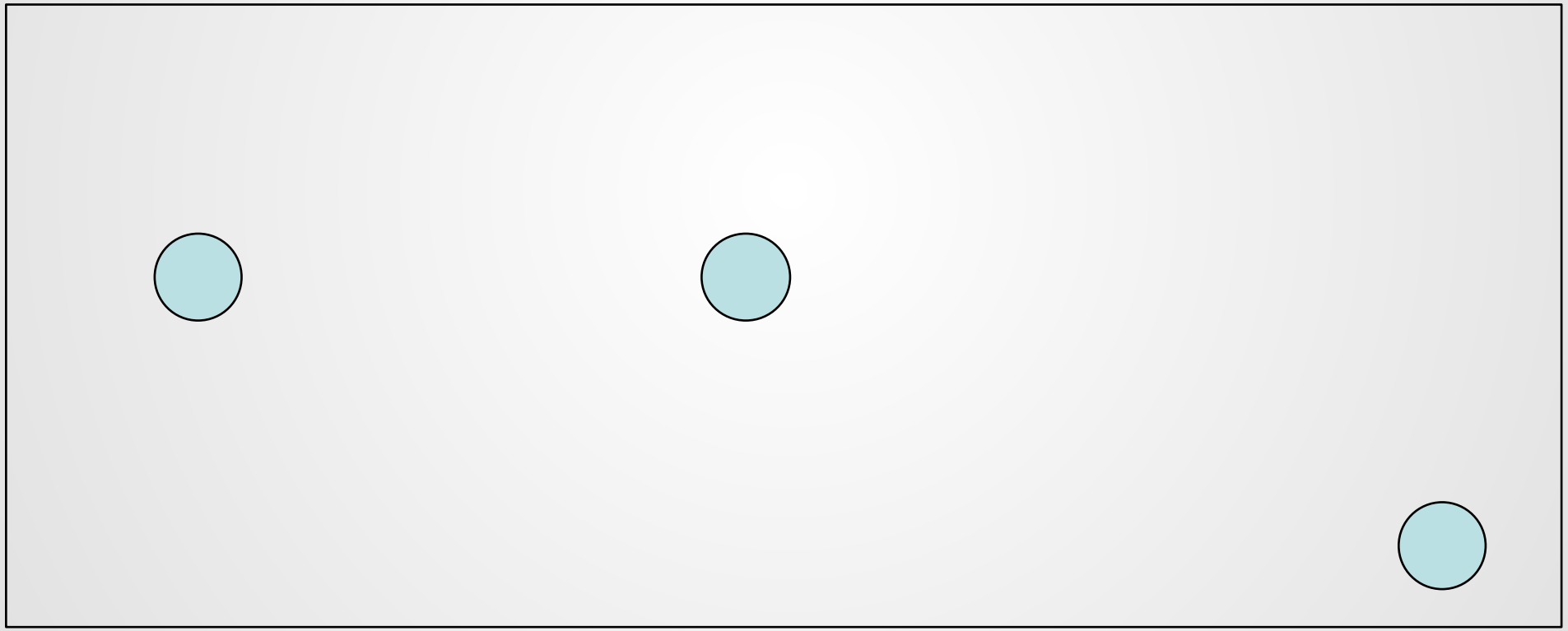
Range and Bearing Measurement Example

- *Linear Example*
- *Range-Bearing Measurement Example*
- *Range-Only Measurement Examples*
- *Range and Bearing Measurement Example*
- *Simple Time-Based Estimation Example*

Range and Bearing Measurement Example

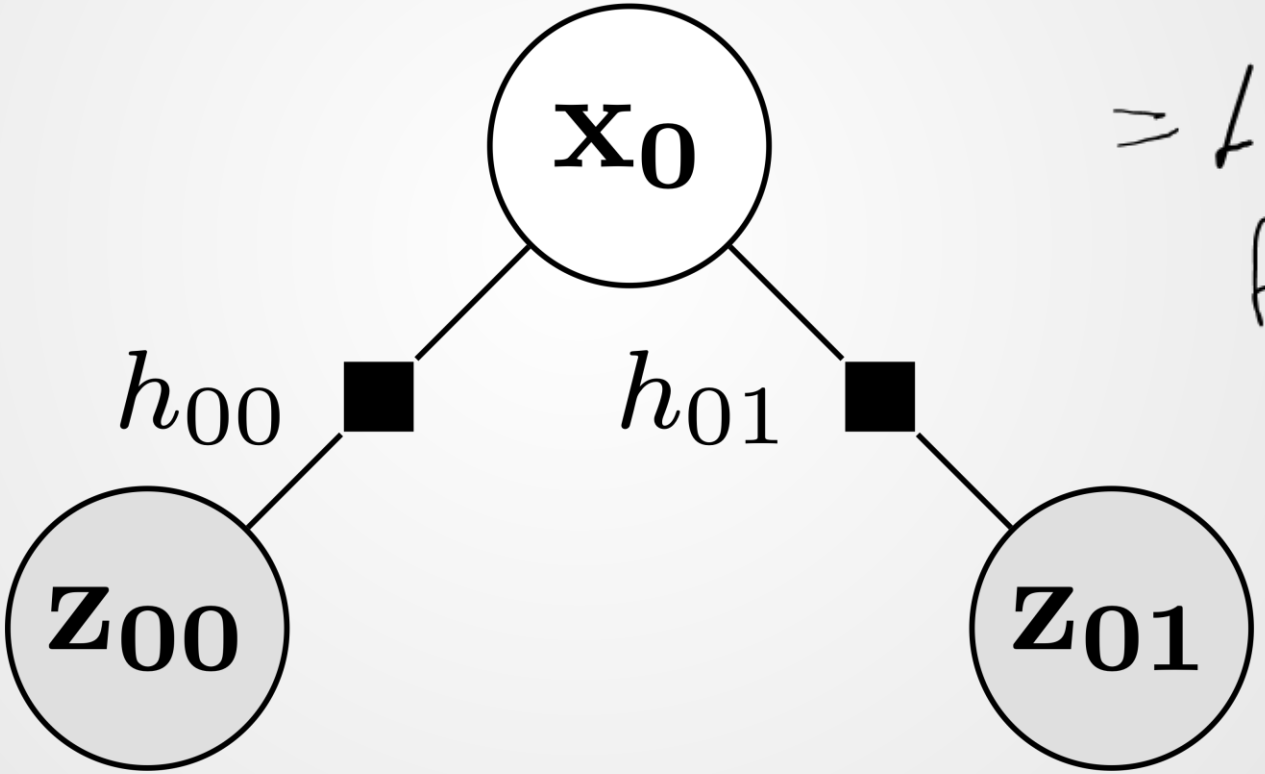
- In this example, we observe the target with two different sensors at two different locations:
  - A bearing sensor
  - A range sensor

Range and Bearing Measurement Example

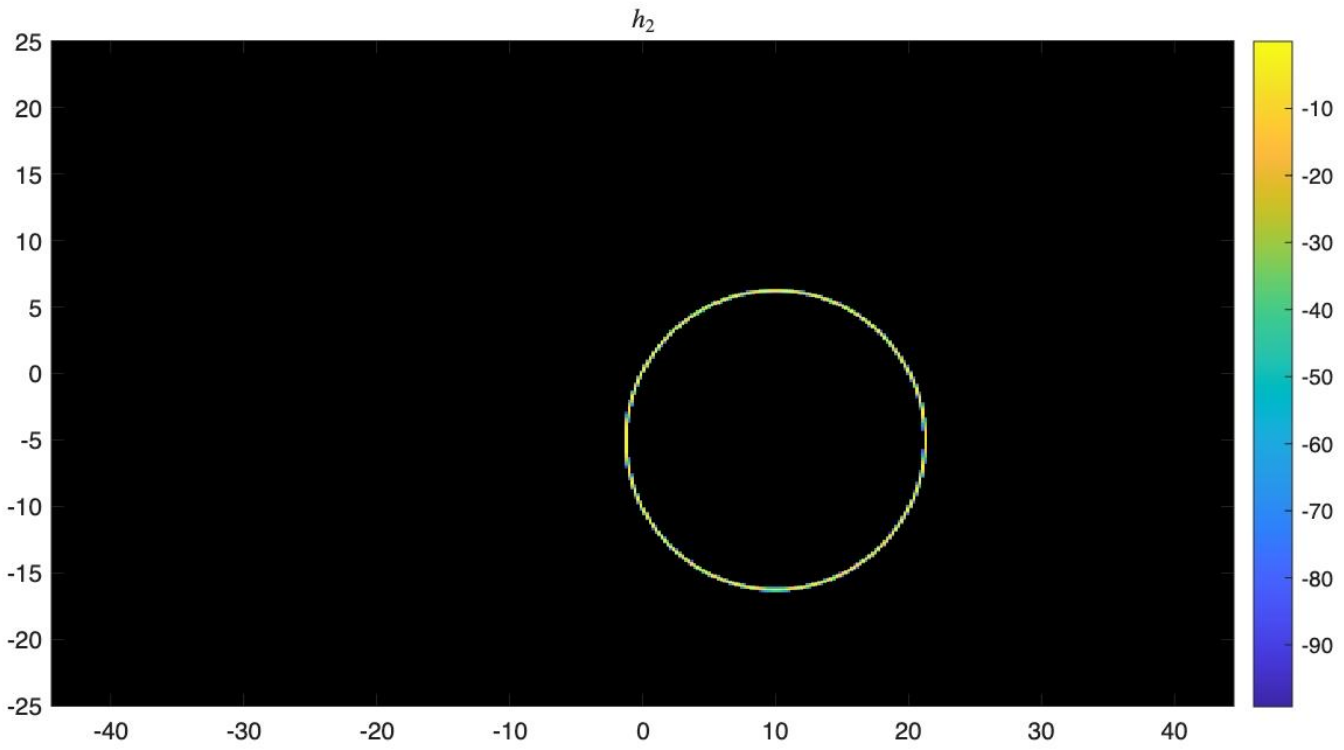


Range and Bearing Measurements

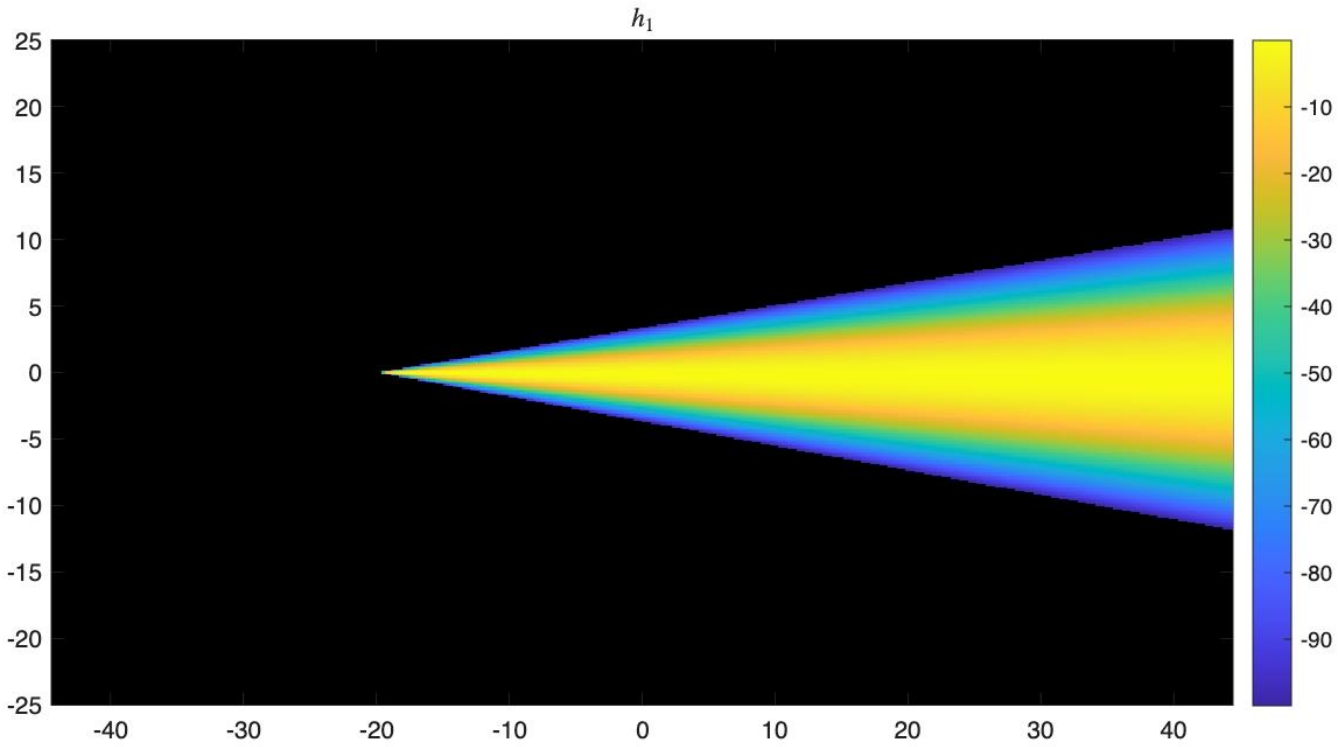
$$f(z_1, z_2 | x) = f(z_1 | x) \, f(z_2 | x)$$



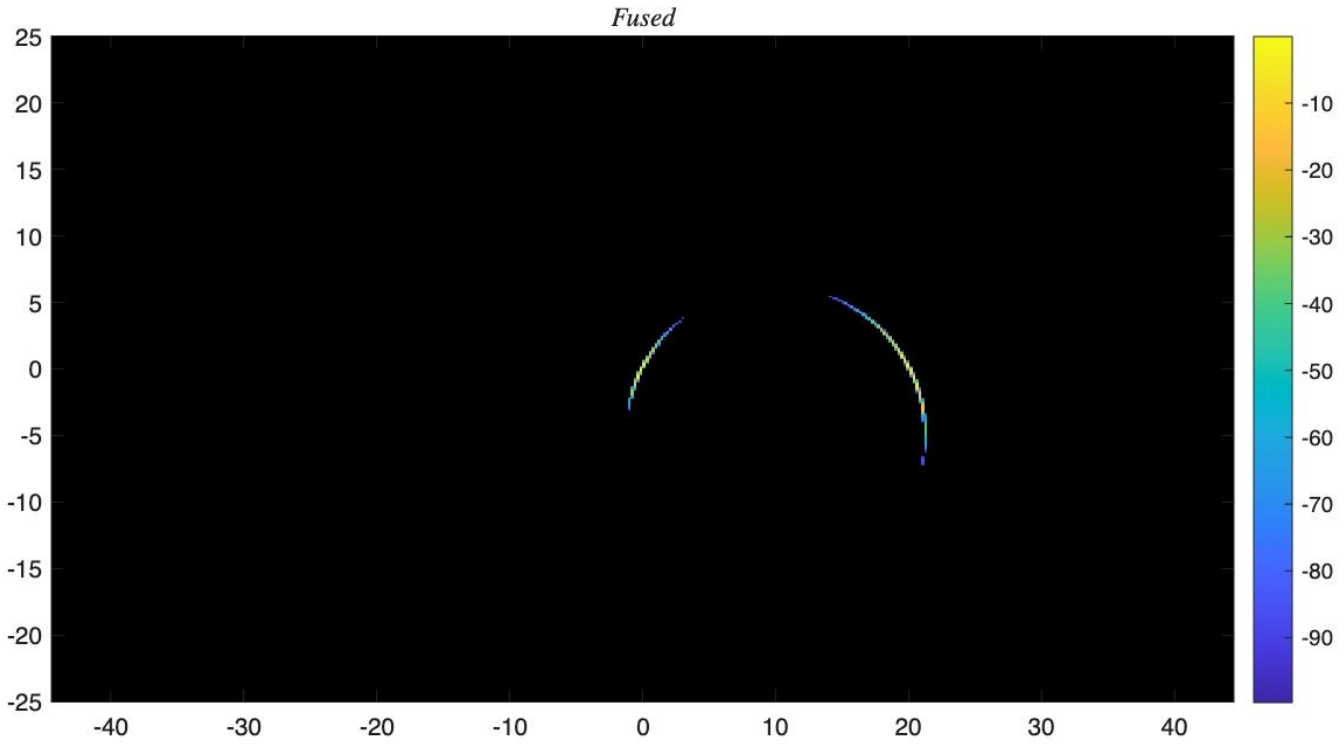
Range Likelihood



Bearing Likelihood



Fused Estimate

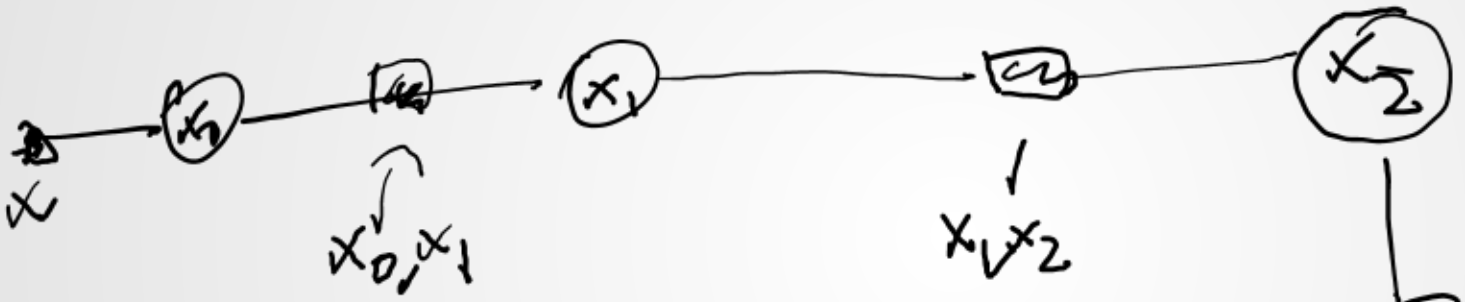


Simple Time-Based Estimation Example

- *Linear Example*
- *Range-Bearing Measurement Example*
- *Range-Only Measurement Examples*
- *Angle-Only Measurement Examples*
- **Simple Time-Based Estimation Example**

$$x_{k+1} = f[x_k, u_{k+1}] + v_{k+1}$$

$$\exp\left(-\frac{1}{2} e^T Q^{-1} e\right)$$



$x_{i|j}$

x_{k}, z_2

$$e[x_k, x_{k+1}, u_{k+1}] = x_{k+1} - f(x_k, u_{k+1})$$

z_2

Smoothing

Summary

- We’ve looked at a few examples of what factor graphs look like
- We have focused on how the observations and measurement likelihoods work through
- We’ve seen that even with Gaussian noises, we can get non-Gaussian likelihoods
- These can reflect things like multi-modality, skewness, etc.
- However, this just tells us the distributions encoded in the graph - how do we extract a state estimate?